



模式识别与计算机视觉

Chapter 2: Lighting and Color

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Overview



- **What is an image?**
- Reflectance properties of surfaces
- What is color: Physical origin of color
- What is color: Physiology of color
- Quantifying color perception and color space
- Color constancy



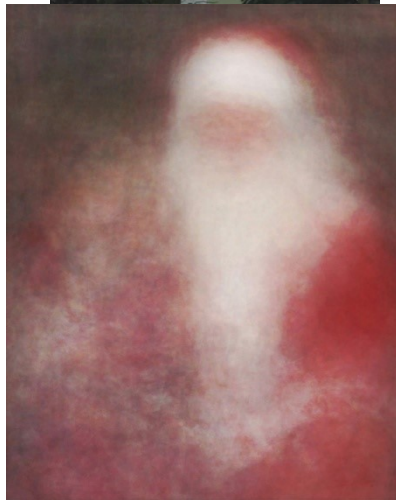
Images as matrices



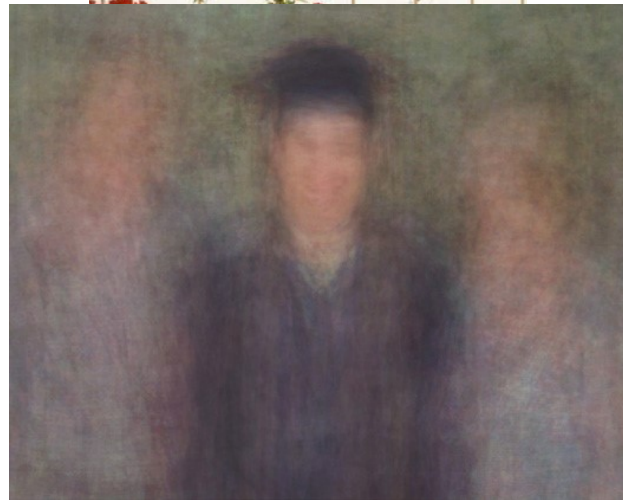
Result of averaging 100 similar snapshots



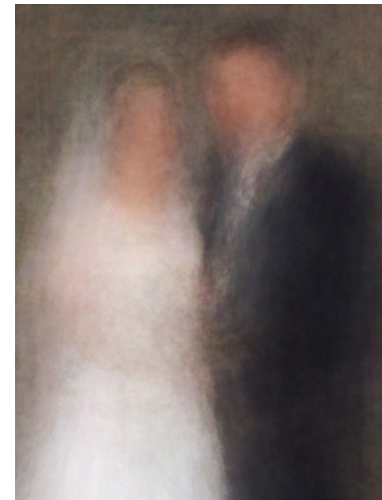
Little Leaguer



Kids with Santa



The Graduate

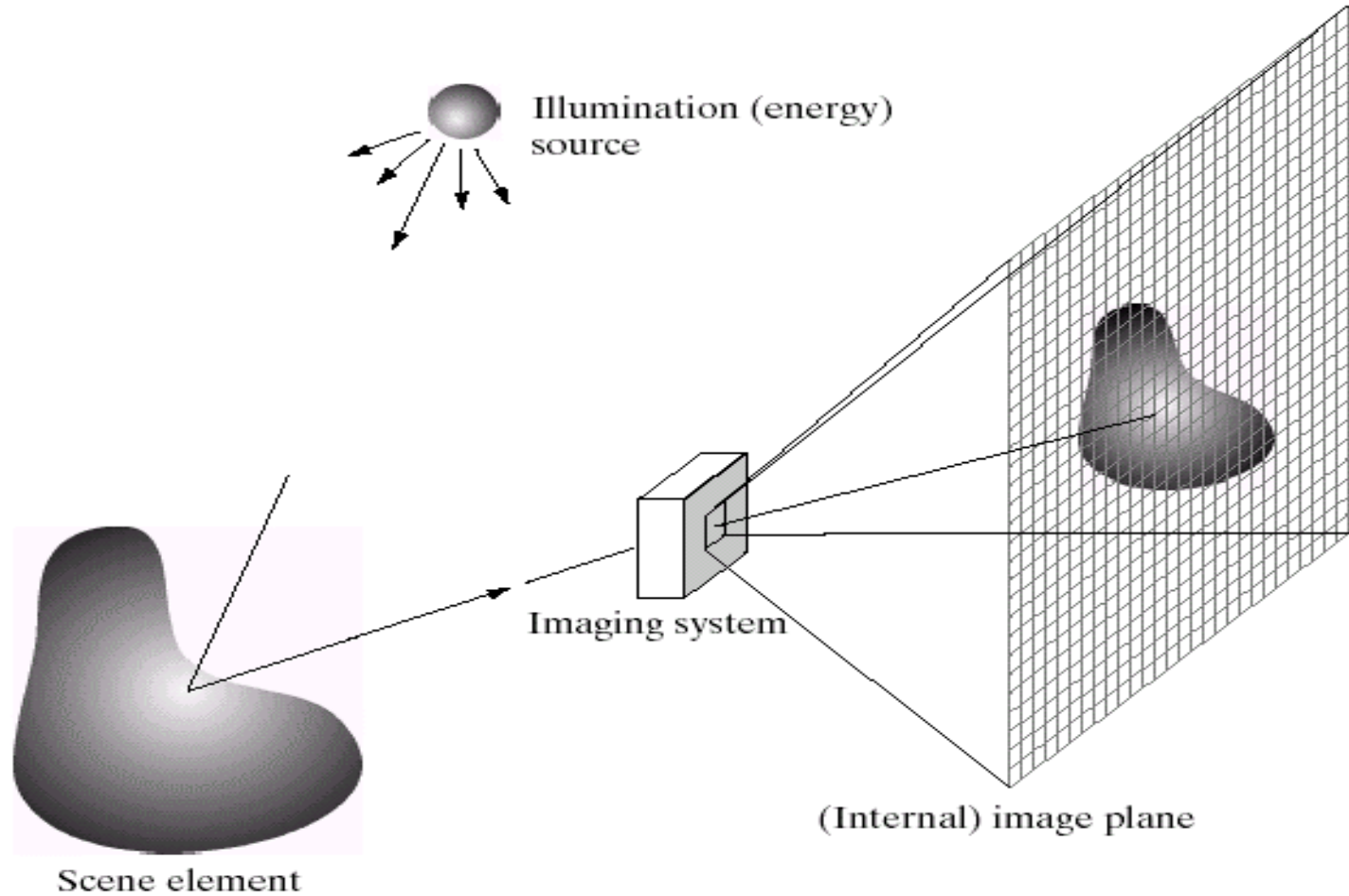


Newlyweds

From: *100 Special Moments*, by Jason Salavon (2004)
<http://salavon.com/SpecialMoments/SpecialMoments.shtml>



How light is recorded





Digital camera

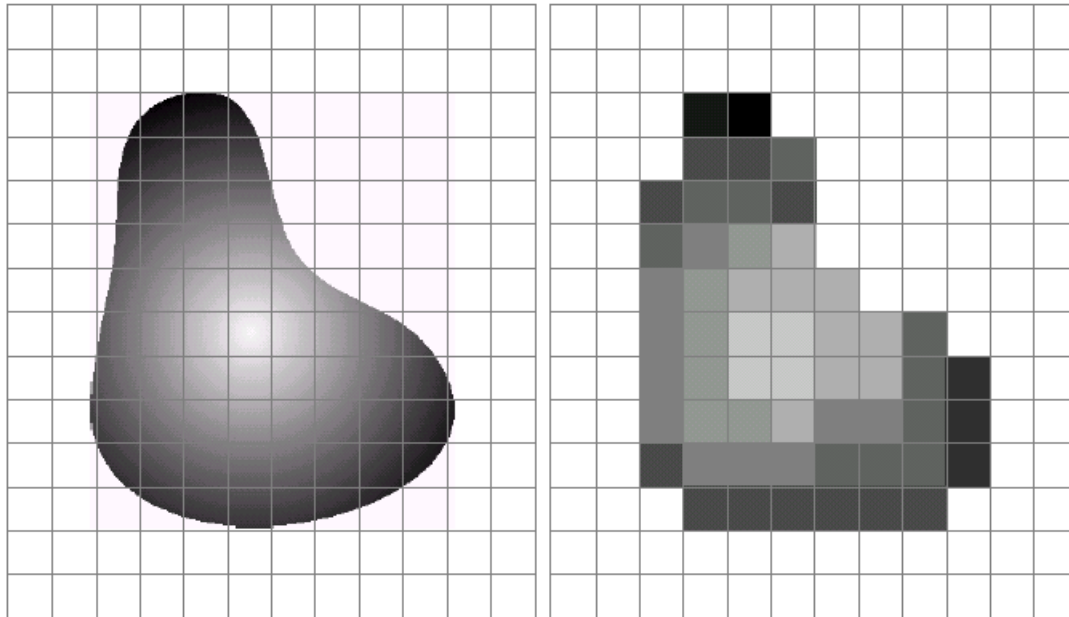


A digital camera replaces film with a sensor array

- Each cell in the array is light-sensitive diode that converts photons to electrons
- <http://electronics.howstuffworks.com/digital-camera.htm>



Sensor array



a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.



CMOS sensor

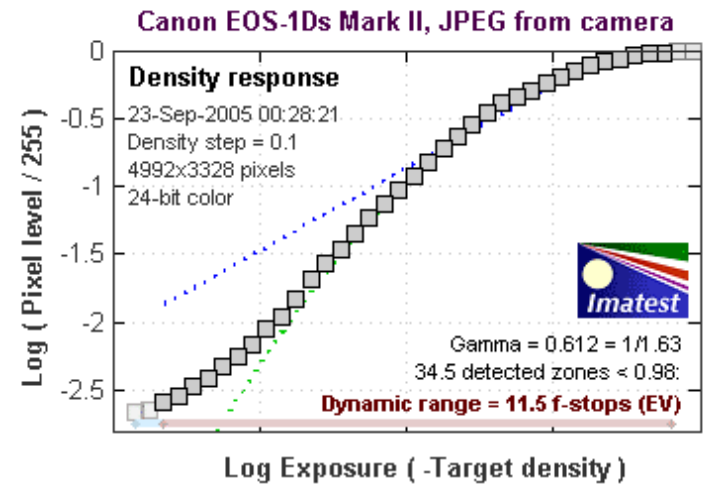
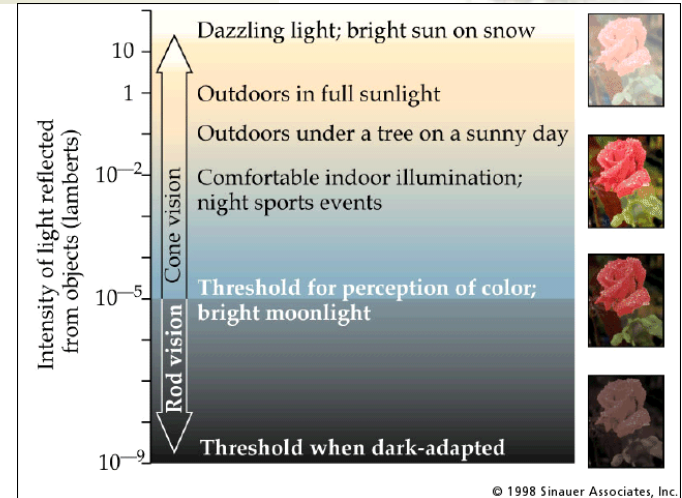
Each sensor cell records amount of light coming in at a small range of orientations



Dynamic range and camera response

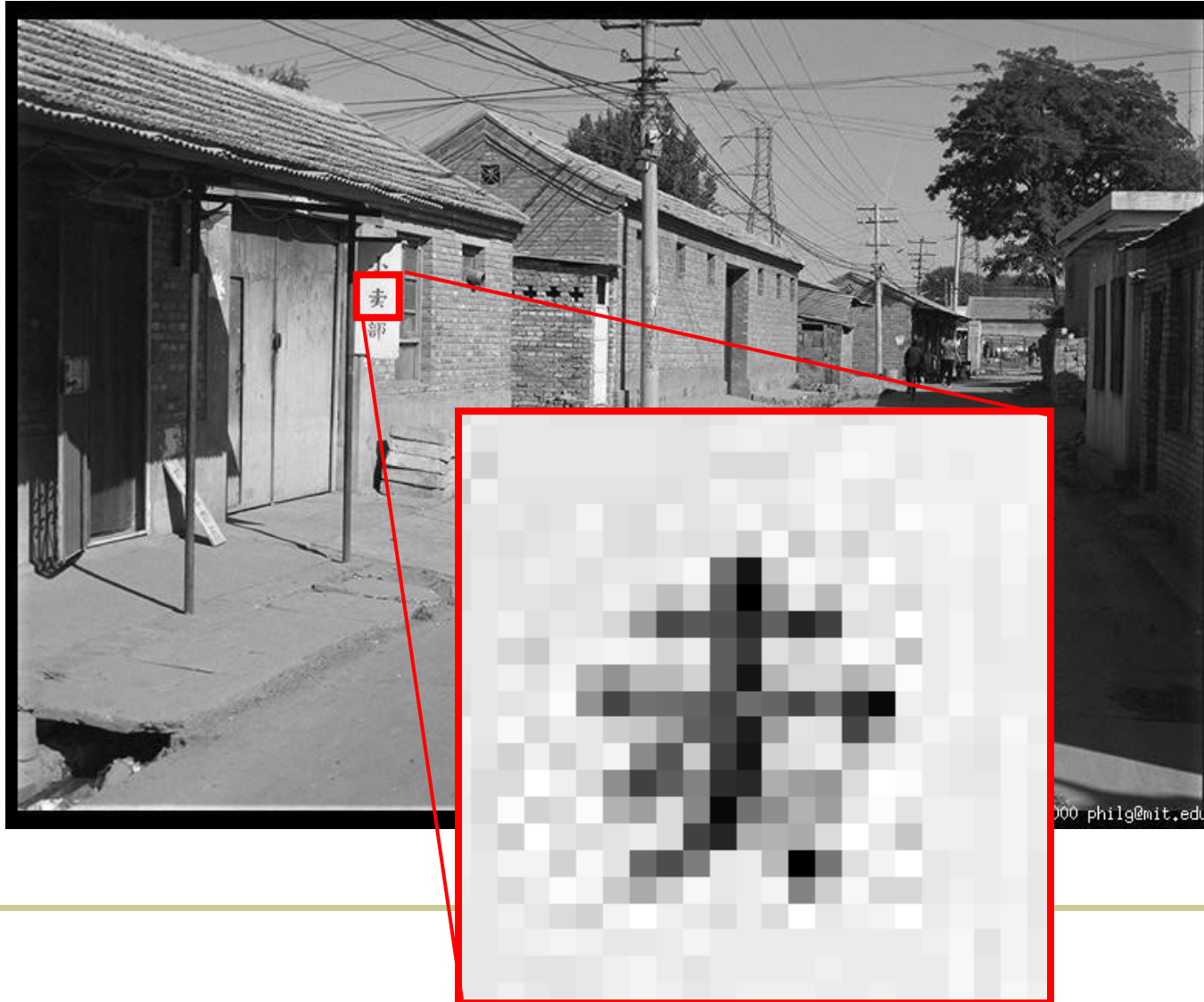


- Typical scenes have a huge dynamic range
- Camera response is roughly linear in the mid range (15 to 240) but non-linear at the extremes
 - called saturation or undersaturation





The raster image (pixel matrix)

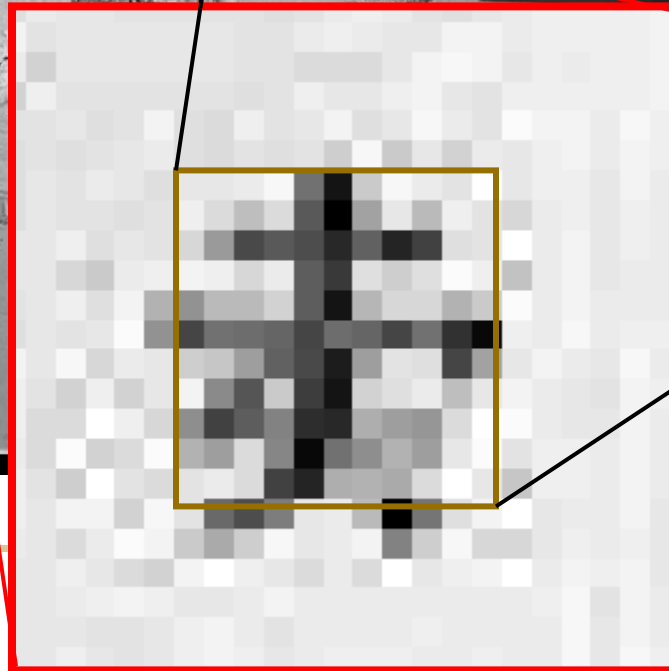




The raster image (pixel matrix)



0.92	0.93	0.94	0.97	0.62	0.37	0.85	0.97	0.93	0.92	0.99
0.95	0.89	0.82	0.89	0.56	0.31	0.75	0.92	0.81	0.95	0.91
0.89	0.72	0.51	0.55	0.51	0.42	0.57	0.41	0.49	0.91	0.92
0.96	0.95	0.88	0.94	0.56	0.46	0.91	0.87	0.90	0.97	0.95
0.71	0.81	0.81	0.87	0.57	0.37	0.80	0.88	0.89	0.79	0.85
0.49	0.62	0.60	0.58	0.50	0.60	0.58	0.50	0.61	0.45	0.33
0.86	0.84	0.74	0.58	0.51	0.39	0.73	0.92	0.91	0.49	0.74
0.96	0.67	0.54	0.85	0.48	0.37	0.88	0.90	0.94	0.82	0.93
0.69	0.49	0.56	0.66	0.43	0.42	0.77	0.73	0.71	0.90	0.99
0.79	0.73	0.90	0.67	0.33	0.61	0.69	0.79	0.73	0.93	0.97
0.91	0.94	0.89	0.49	0.41	0.78	0.78	0.77	0.89	0.99	0.93



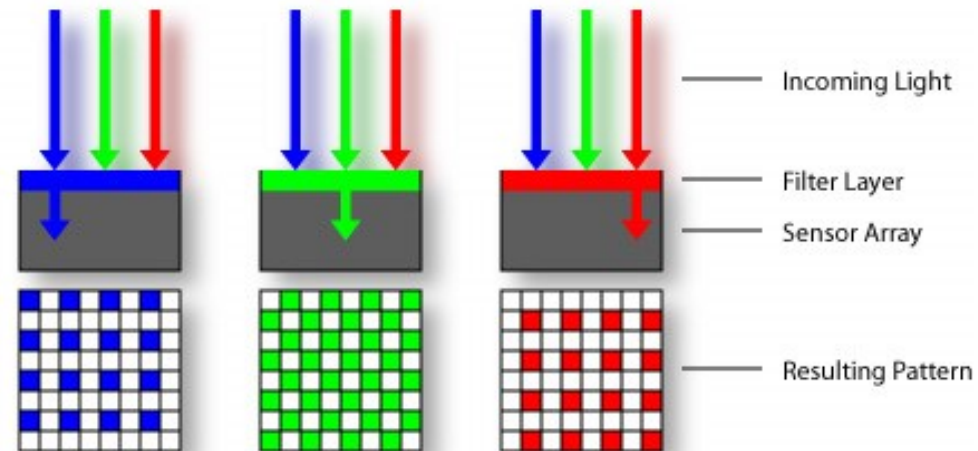
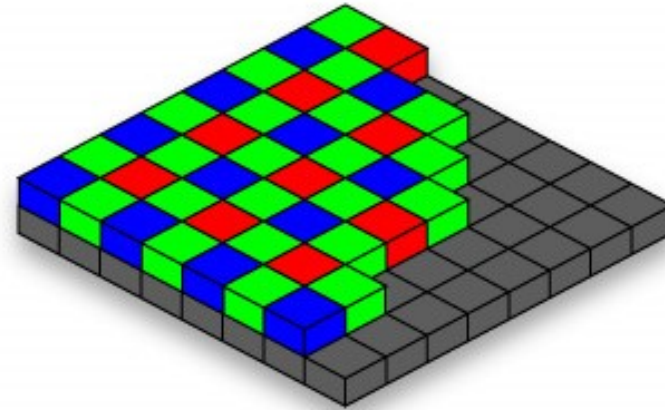
000 philg@mit.edu



Color Sensing: Bayer Grid



- Estimate RGB at each cell from neighboring values





Color Image



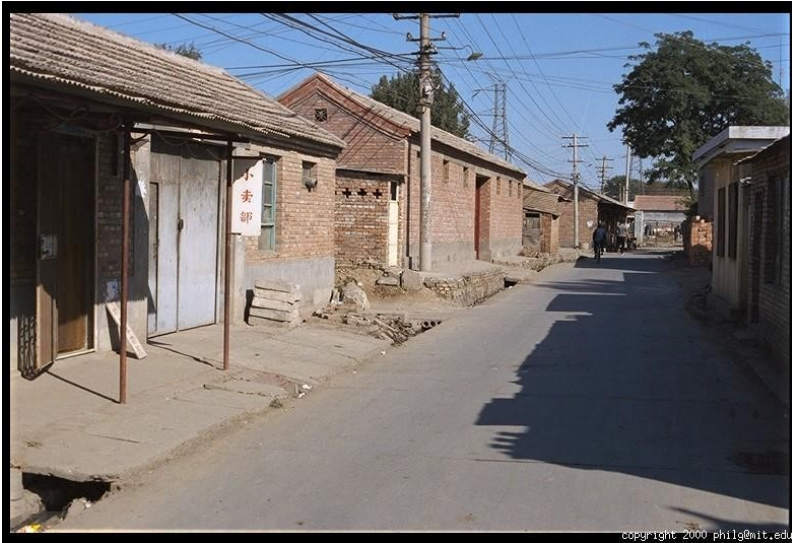
G



B



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Overview



- What is an image?
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What determines a pixel's intensity



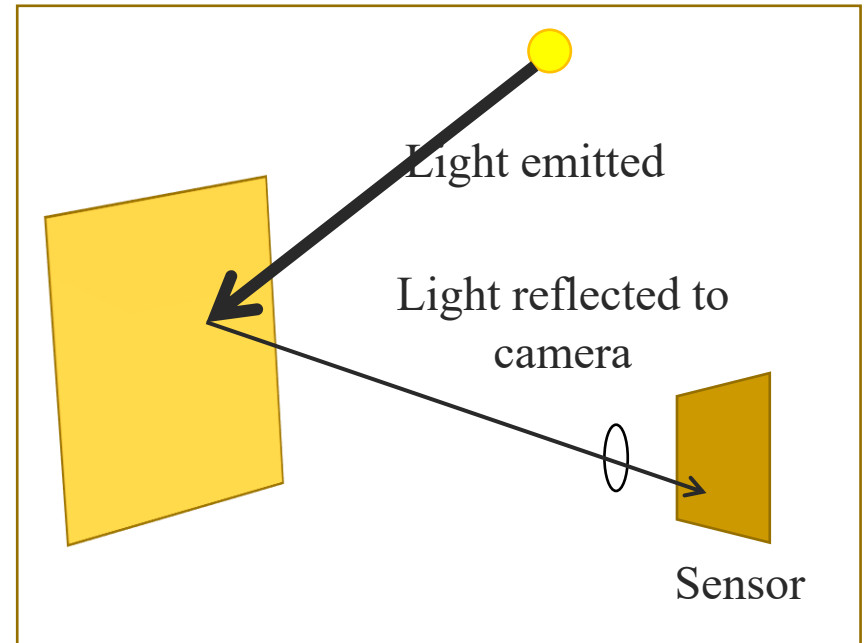


How much light is recorded



■ Major factors

- Illumination strength and direction
- Surface geometry
- Surface material
- Nearby surfaces
- Camera gain/exposure

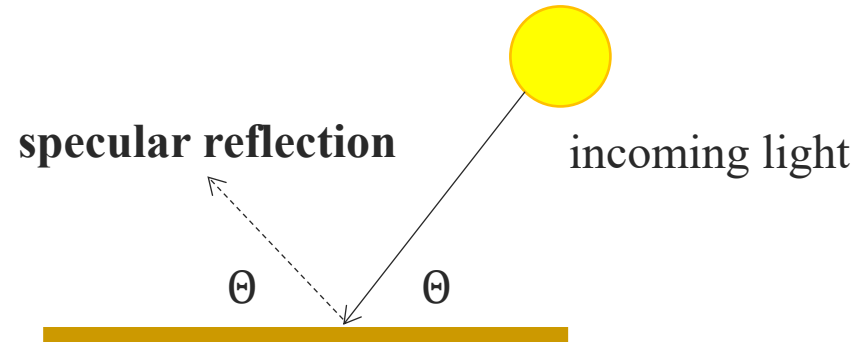




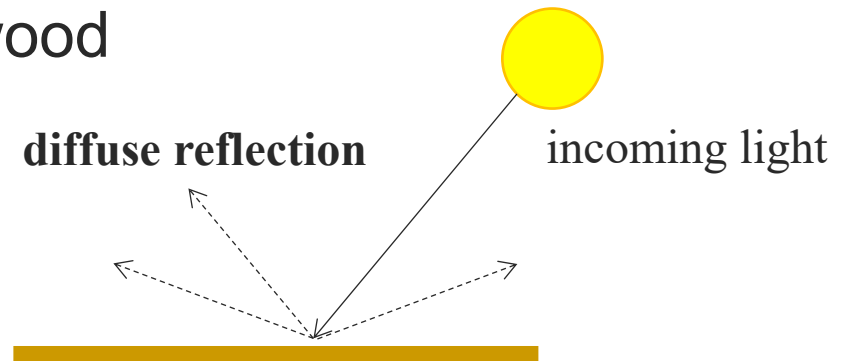
Basic models of reflection



- Specular: light bounces off at the incident angle
 - E.g., mirror



- Diffuse: light scatters in all directions
 - E.g., brick, cloth, rough wood



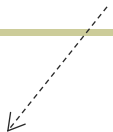
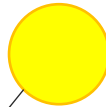


Other possible effects



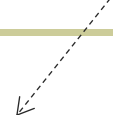
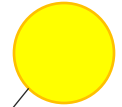
Transparency (透视)

light source



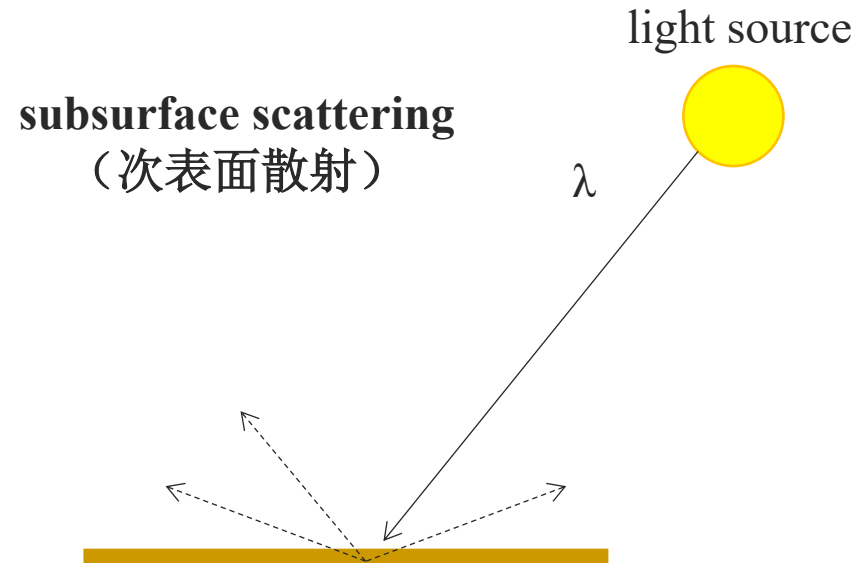
Refraction (折射)

light source



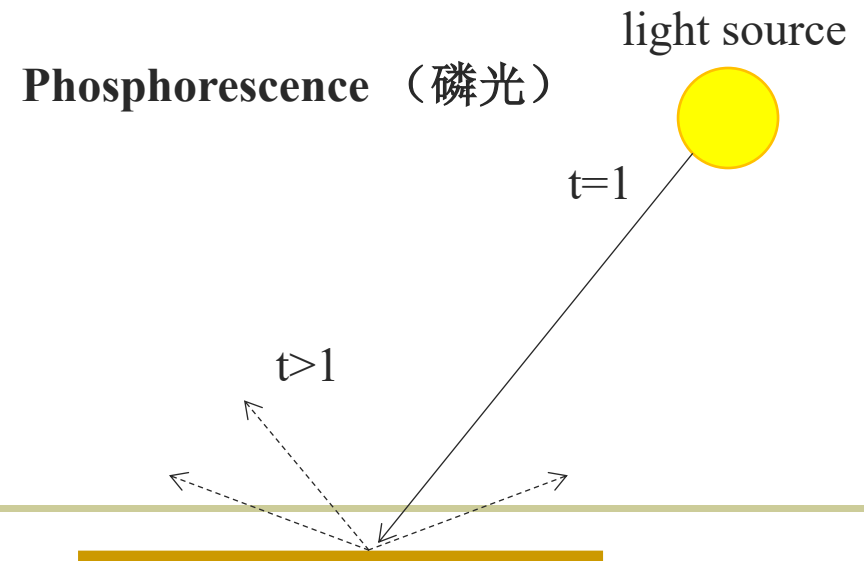
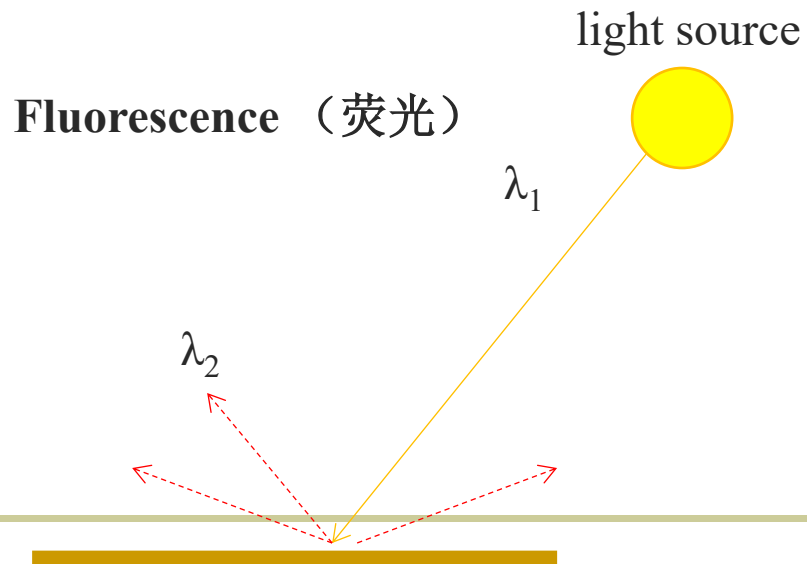


Other possible effects





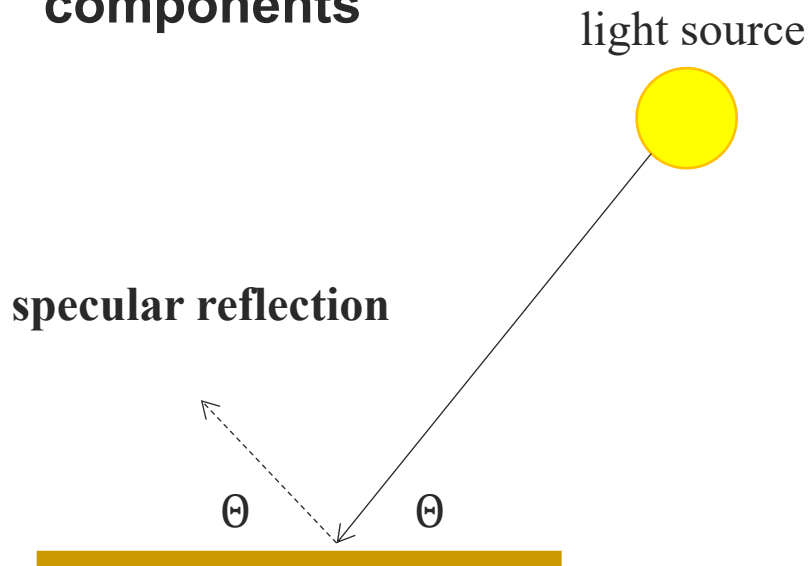
Other possible effects





Specular Reflection (镜面反射)

- Reflected direction depends on light orientation and surface normal
 - E.g., mirrors are fully specular
 - **Most surfaces can be modeled with a mixture of diffuse and specular components**



Flickr, by suzysputnik



Flickr, by piratejohnny



Most surfaces have both specular and diffuse components



- Specularity (高光) = spot where specular reflection dominates (typically reflects light source)



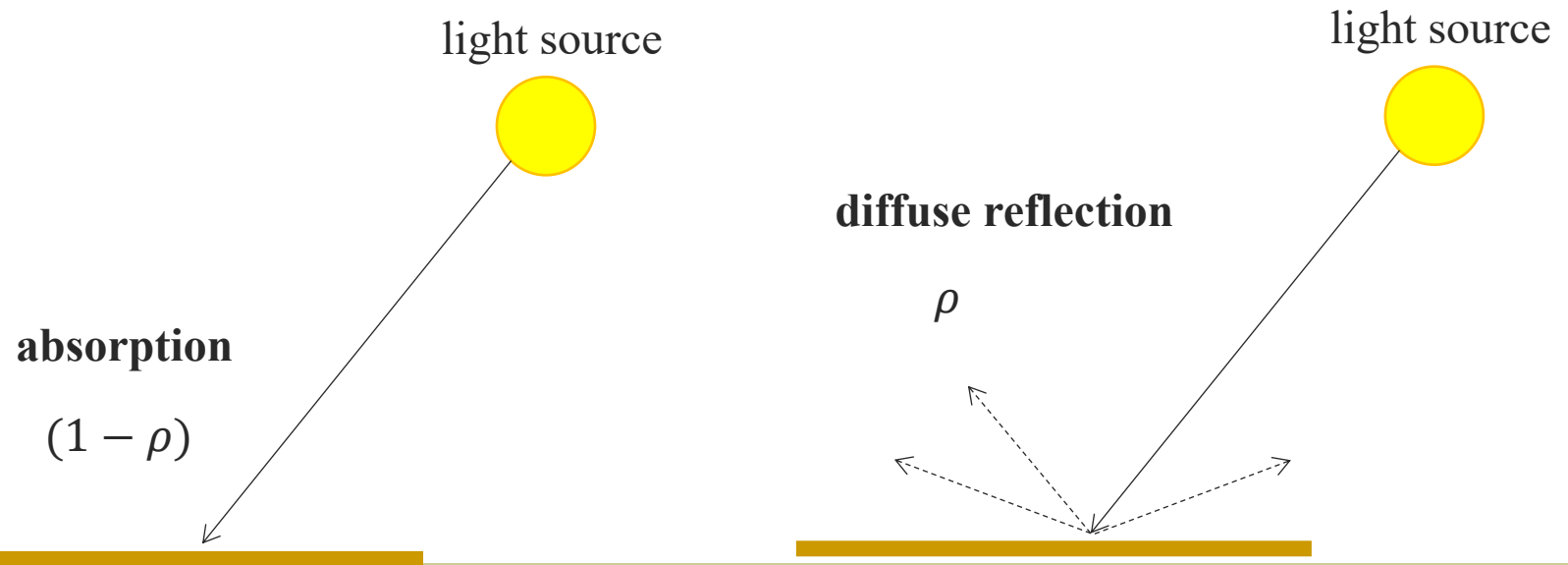
Typically, specular component is small



Lambertian reflectance model (理想散射)



- Some light is absorbed (function of albedo ρ)
- Remaining light is scattered (diffuse reflection)
- Examples: soft cloth, concrete, matte paints





Intensity and Surface Orientation



Intensity depends on illumination angle because less light comes in at oblique angles.

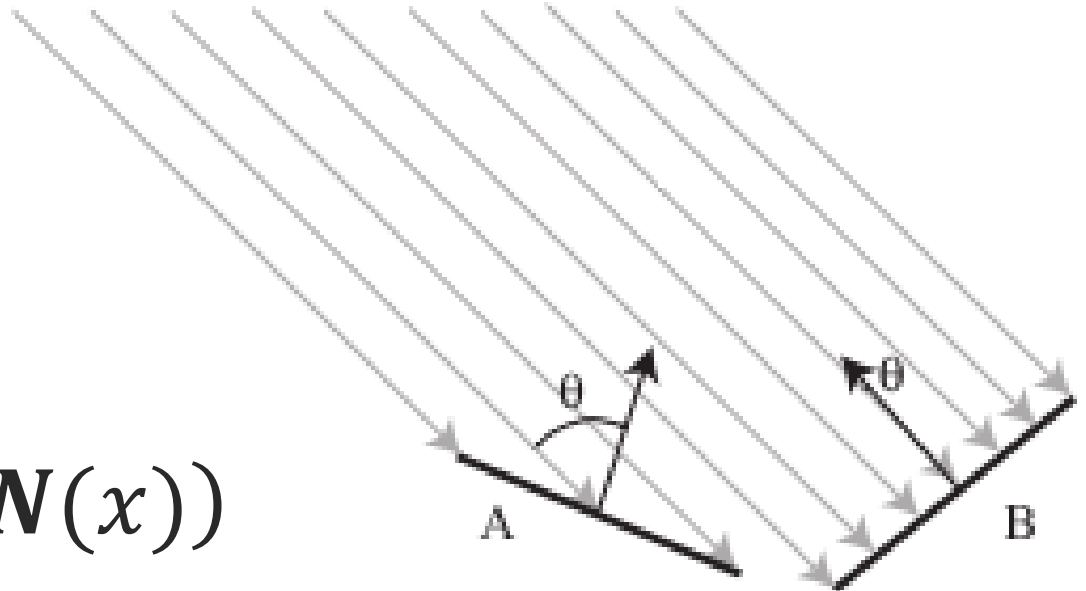
ρ = albedo

$\mathbf{S}$ = directional source

$\mathbf{N}$ = surface normal

I = reflected intensity

$$I(x) = \rho(x)(\mathbf{S} \cdot \mathbf{N}(x))$$



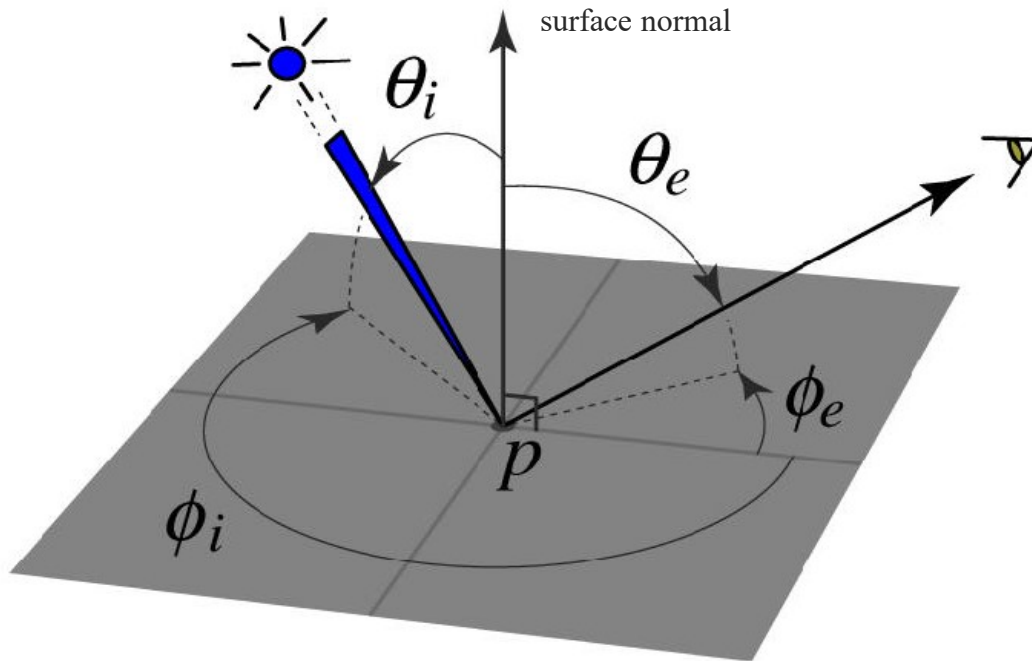




BRDF: Bidirectional Reflectance Distribution Function (双向反射分布函数)



- Model of local reflection
 - How bright a surface appears when viewed from one direction when light falls on it from another

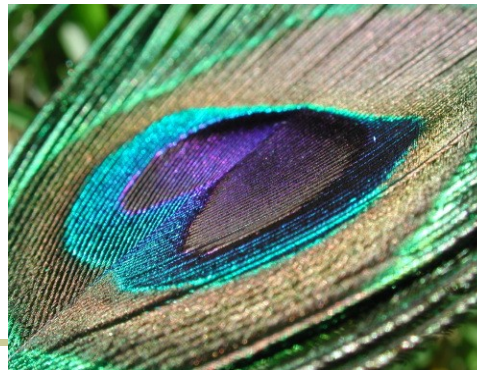


$$\rho(\theta_i, \phi_i, \theta_e, \phi_e; \lambda) =$$

$$\frac{L_e(\theta_e, \phi_e)}{E_i(\theta_i, \phi_i)} = \frac{L_e(\theta_e, \phi_e)}{L_i(\theta_i, \phi_i) \cos \theta_i d\omega}$$



BRDFs can be incredibly complicated

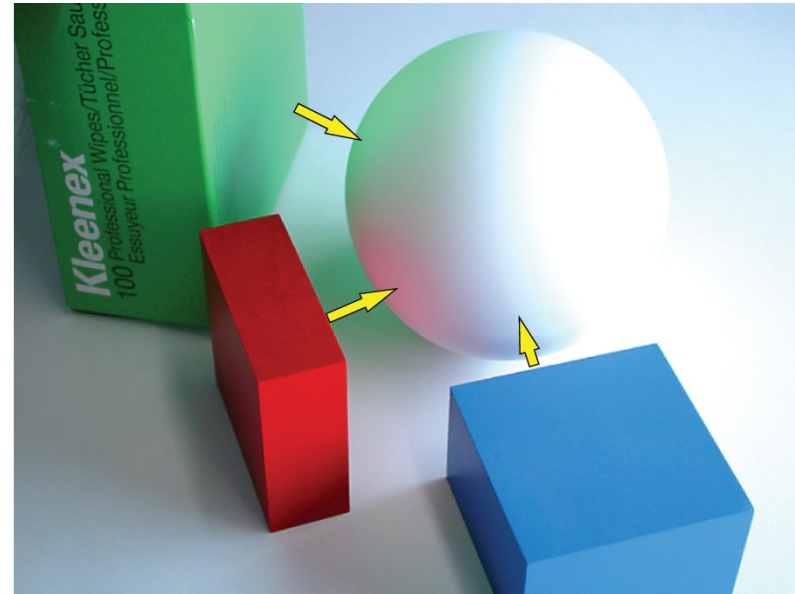




The light of the poor pixel



- A pixel's brightness is determined by
 - Light source (strength, direction, color)
 - Surface orientation
 - Surface material and albedo
 - Reflected light and shadows from surrounding surfaces
 - Gain on the sensor



- A pixel's brightness tells us nothing by itself



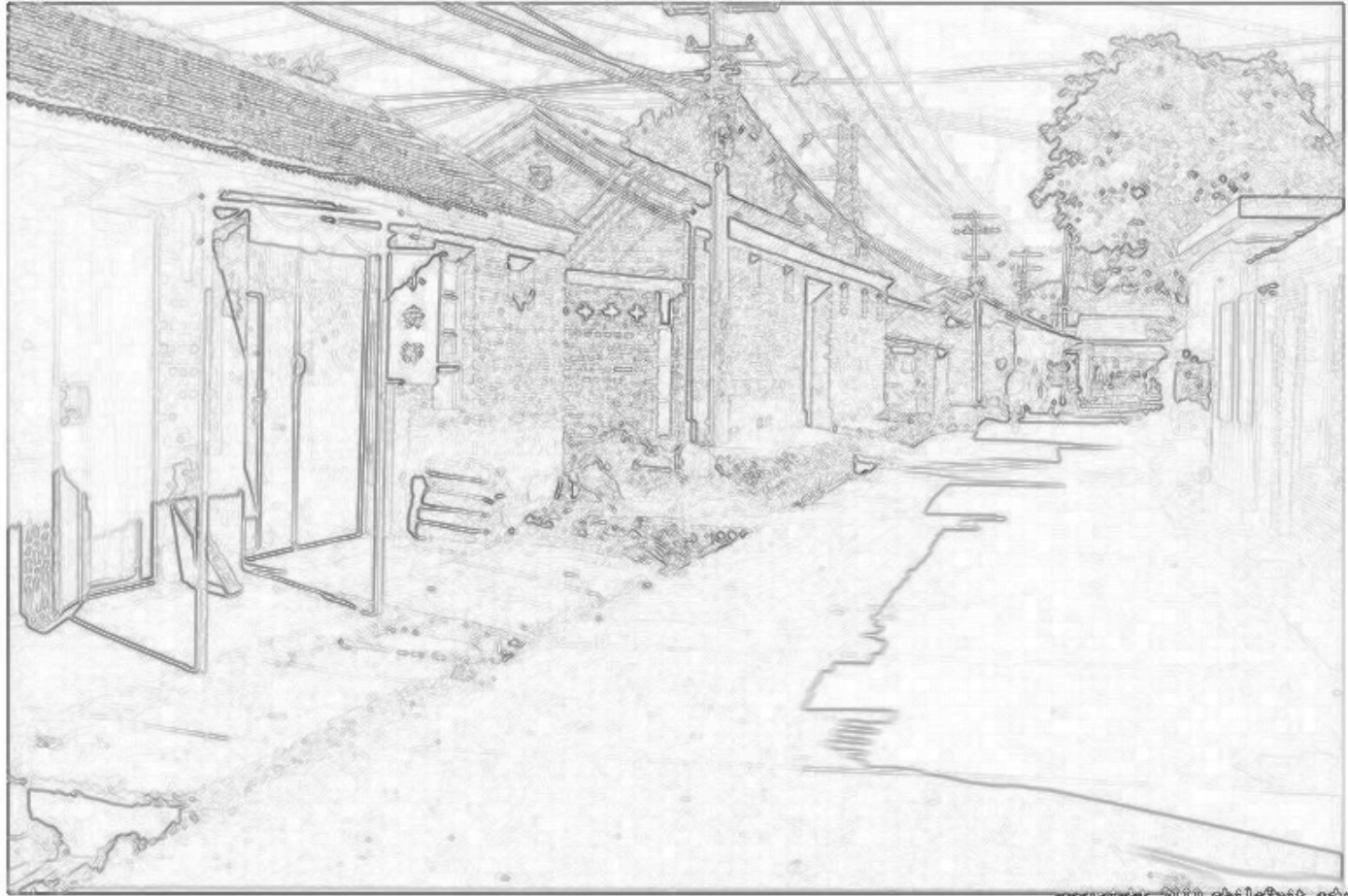
And yet we can interpret images...



- Key idea: for nearby scene points, most factors do not change much
- The information is mainly contained in *local differences* of brightness



Darkness = Large Difference in Neighboring Pixels



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What is this?







Overview



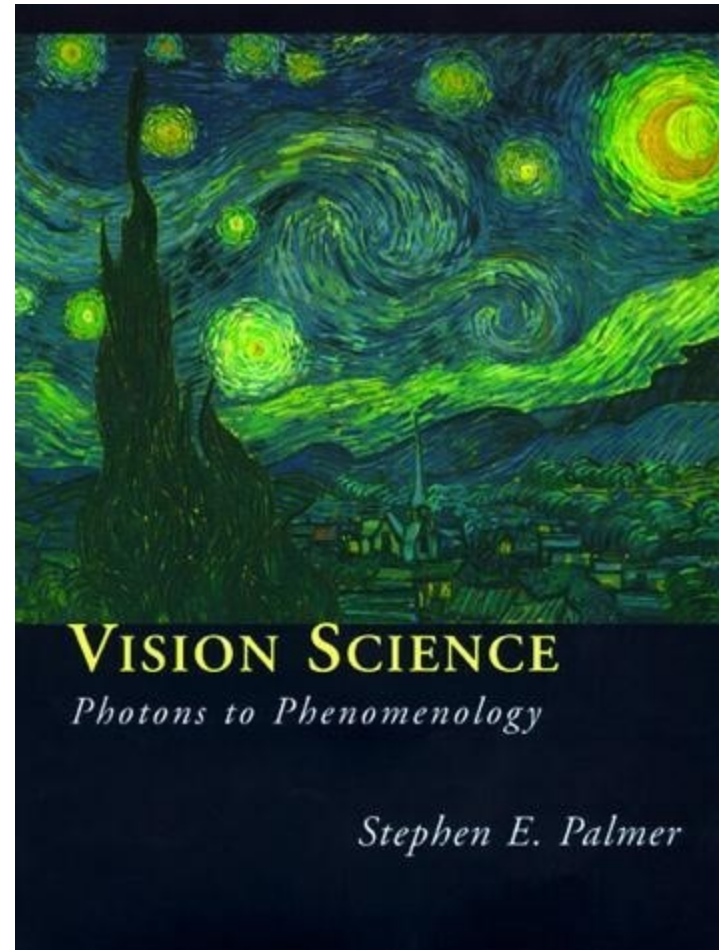
- What is an image?
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What is color?

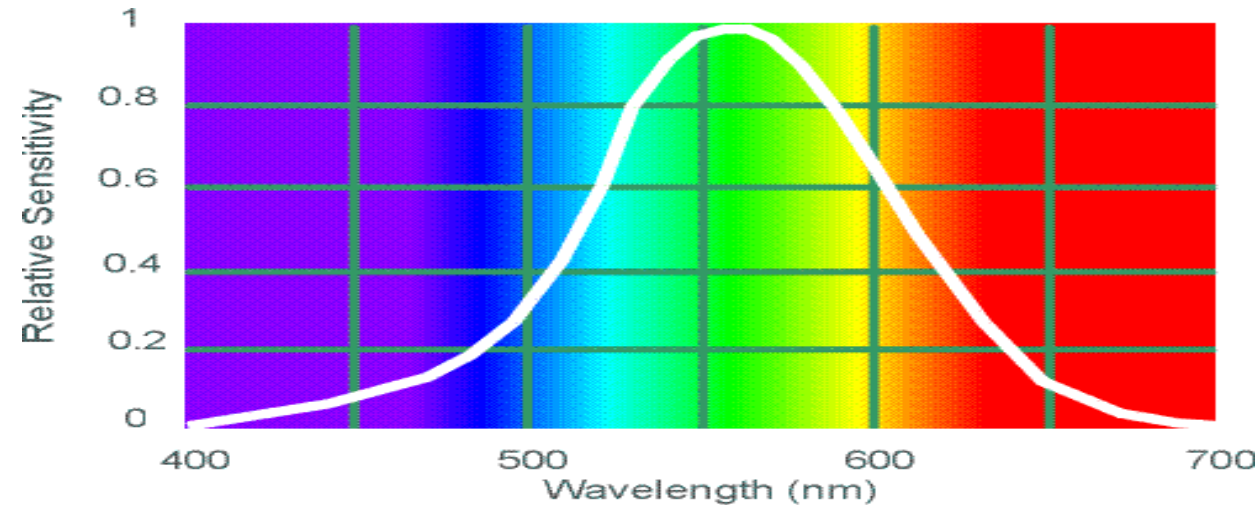
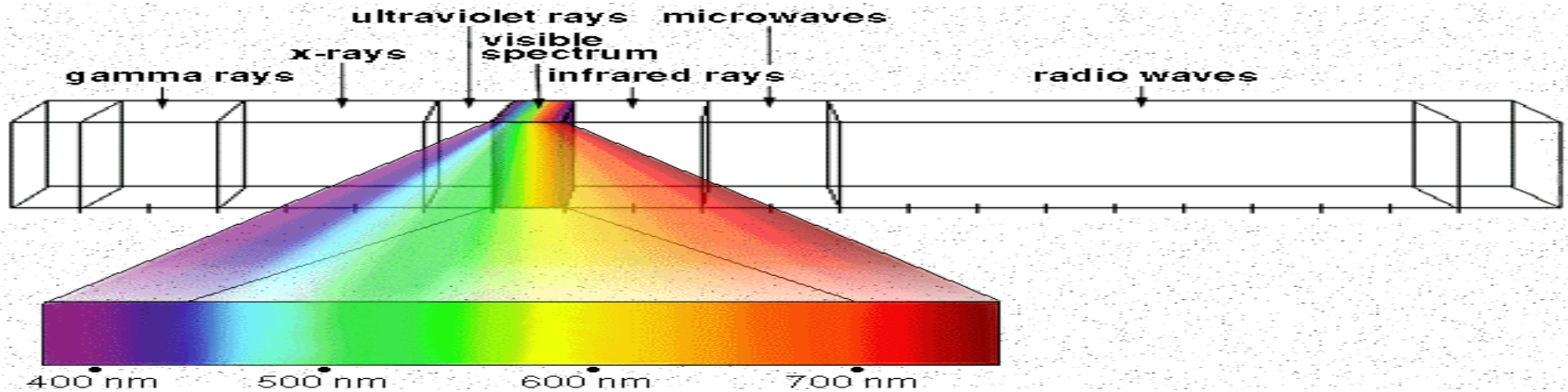


- Color is the result of interaction between **physical light** in the environment and **our visual system**
- Color is a psychological property of our visual experiences when we look at objects and lights, *not* a physical property of those objects or lights
(S. Palmer, *Vision Science: Photons to Phenomenology*)





Color

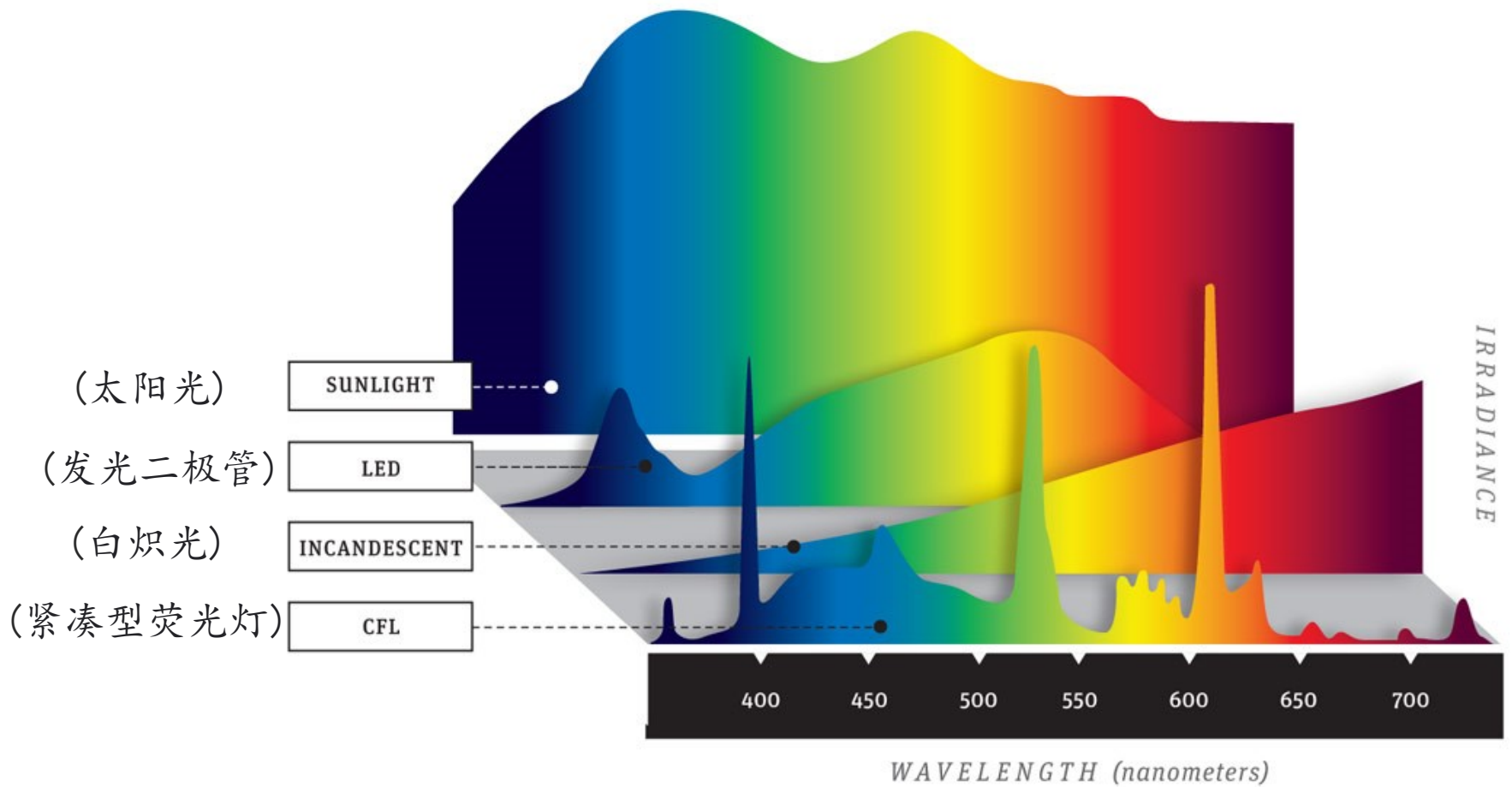


Human Luminance Sensitivity Function

Light is composed
of a spectrum of
wavelengths



Spectra of light sources

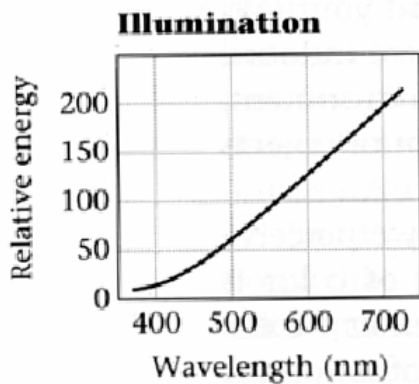




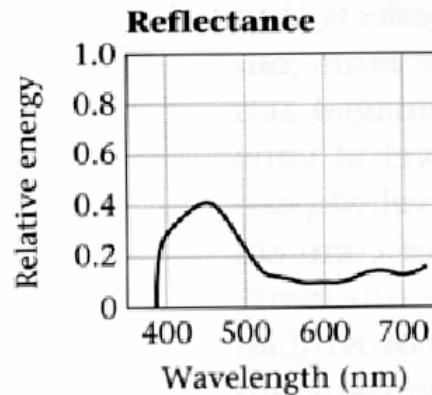
Interaction of light and surfaces



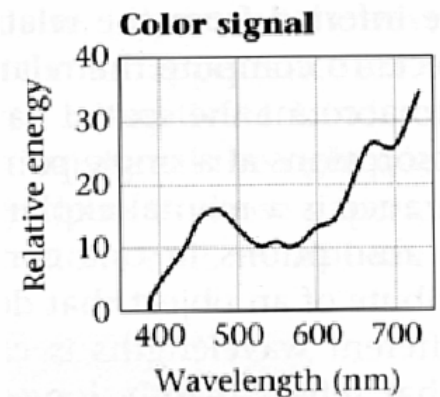
- **Reflected color** is the result of interaction of light source spectrum with surface reflectance



*

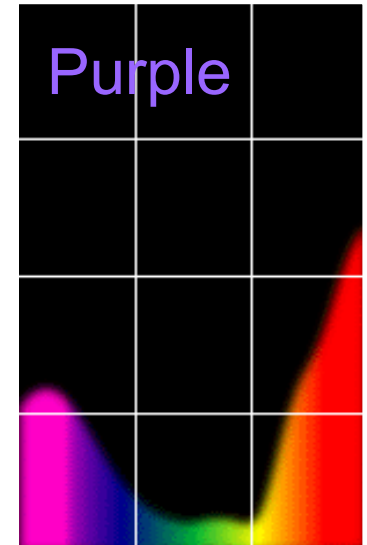
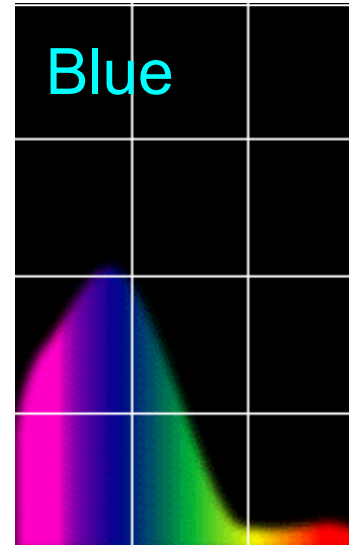
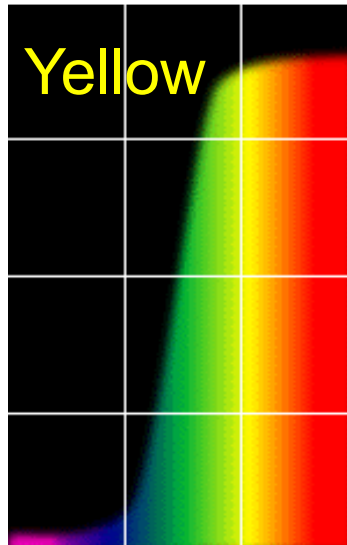
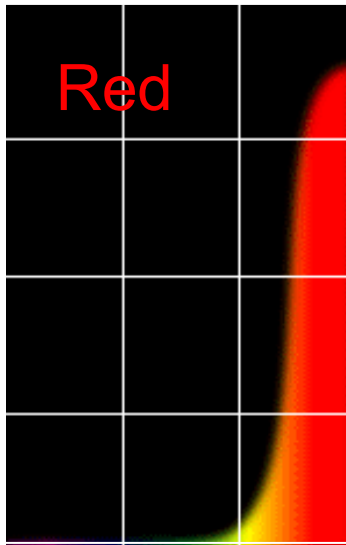


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Some examples of the reflectance spectra of surfaces





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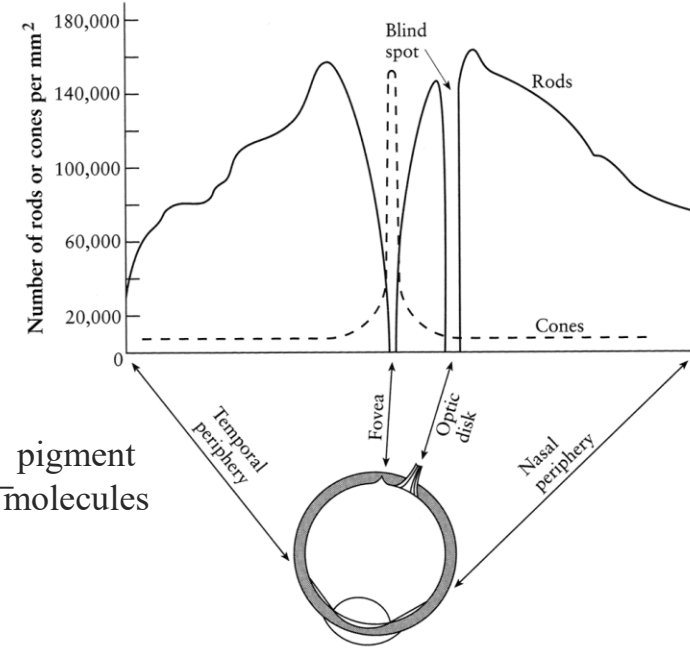
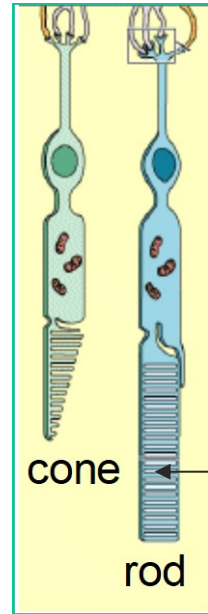
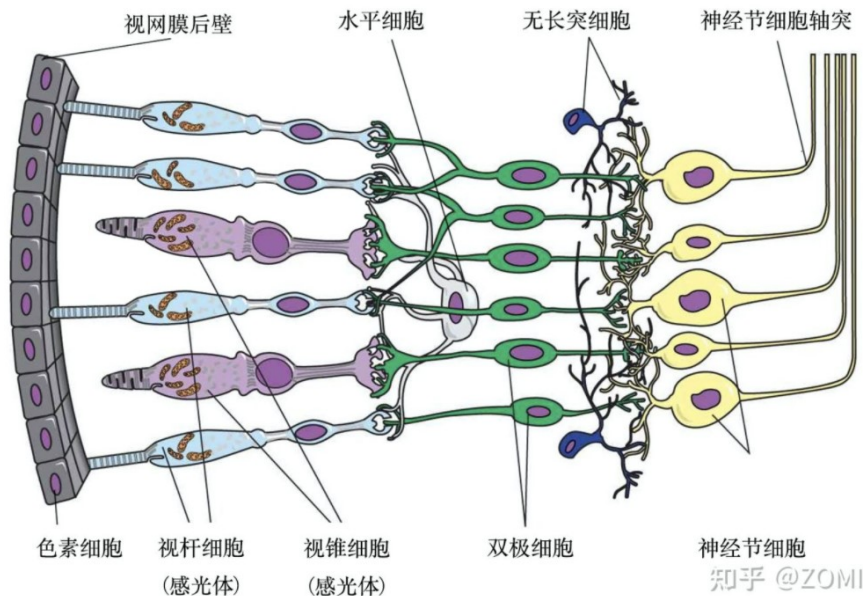
Why RGB?



If light is a spectrum, why are images RGB?



Rods and cones, fovea



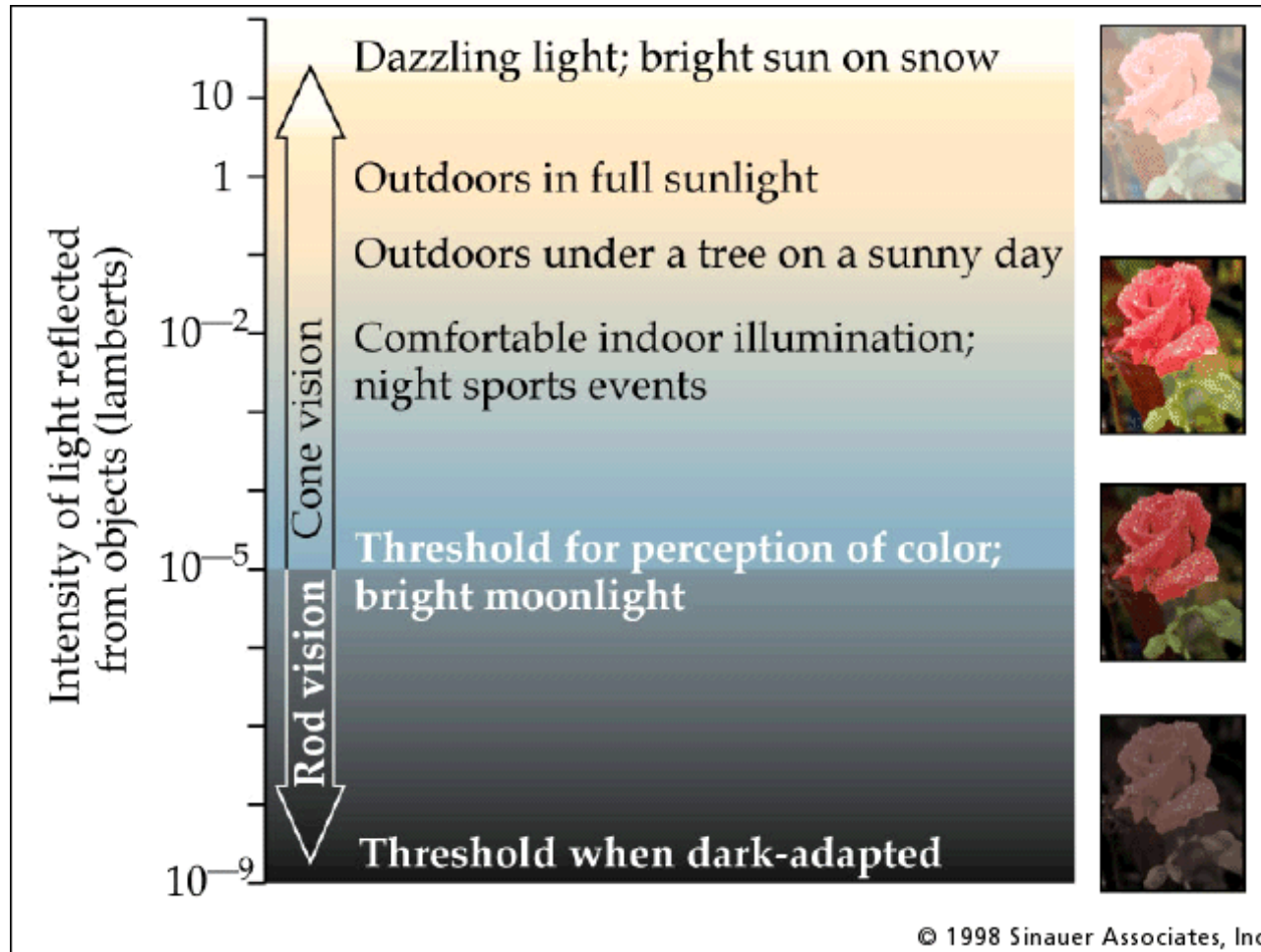
Rods are responsible for intensity, cones for color perception

Rods and cones are *non-uniformly* distributed on the retina

- **Fovea** - Small region (1 or 2°) at the center of the visual field containing the highest density of cones – and no rods

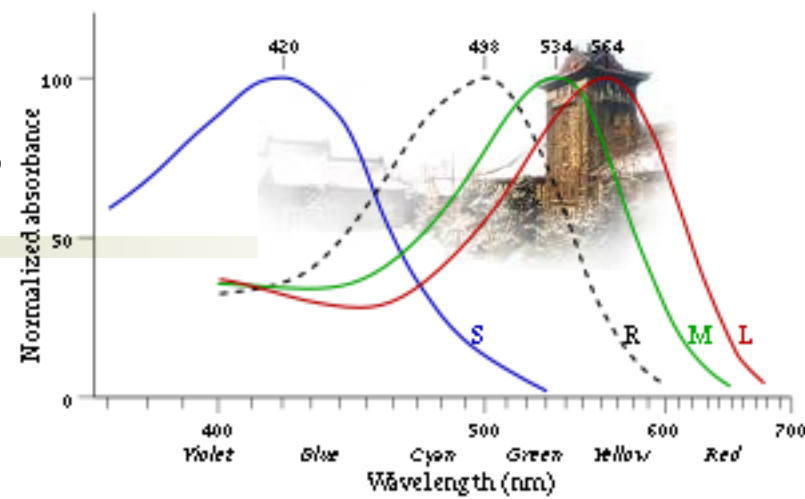


Rod / cone sensitivity





Human color receptors



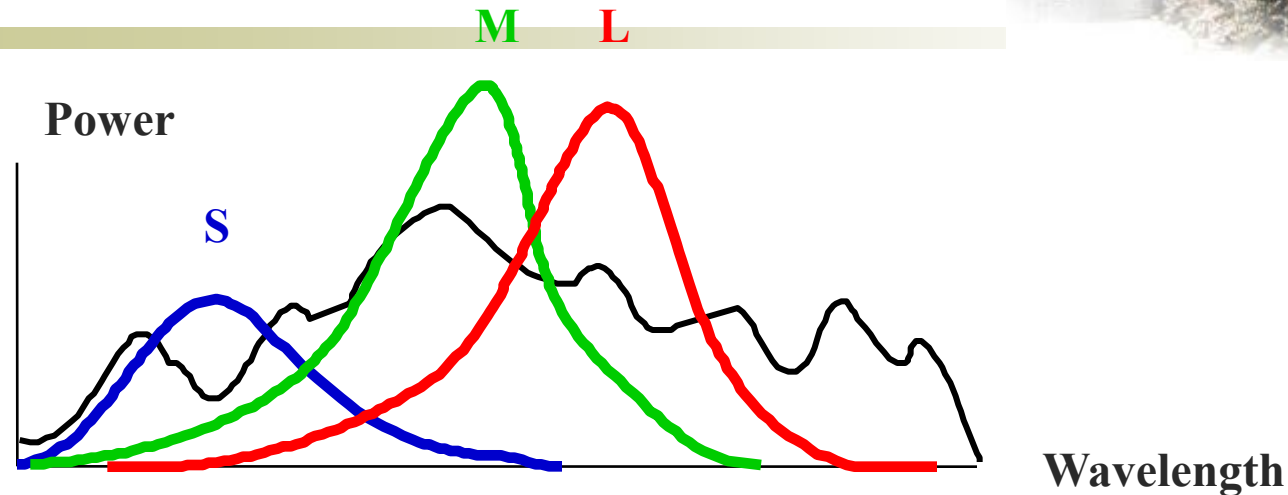
Long (red), Medium (green), and Short (blue) cones, plus intensity rods

■ Fun facts

- “M” and “L” on the X-chromosome
 - That’s why men are more likely to be color blind (see what it’s like:
<http://www.vischeck.com/vischeck/vischeckImage.php>)
- “L” has high variation, so some women are tetrachromatic
- Ratio of L to M to S cones: approx. 10:5:1
- Some animals have 1 (night animals), 2 (e.g., dogs), 4 (fish, birds), 5 (pigeons, some reptiles/amphibians), or even 12 (mantis shrimp) types of cones



Color perception models



1. Rods and cones act as **filters** on the spectrum

- To get the output of a filter, multiply its response curve by the spectrum, integrate over all wavelengths
- Each cone yields one number

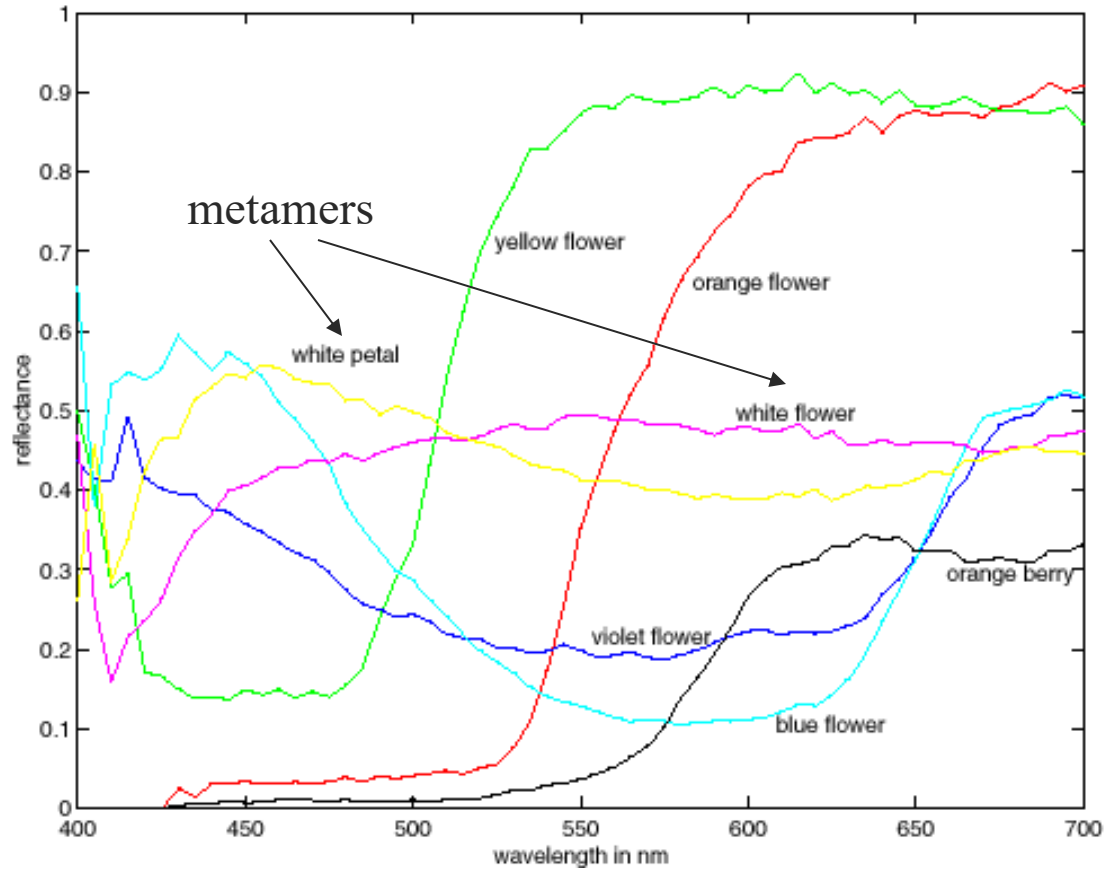
2. How can we represent an entire spectrum with three numbers?

We can't! Most of the information is lost

3. As a result, two different spectra may appear indistinguishable such spectra are known as **metamers** (同色光)

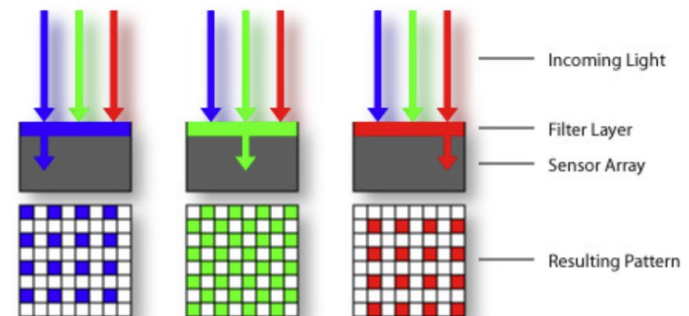
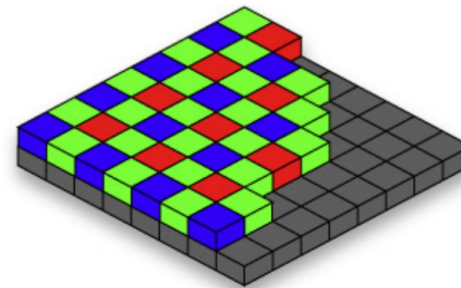
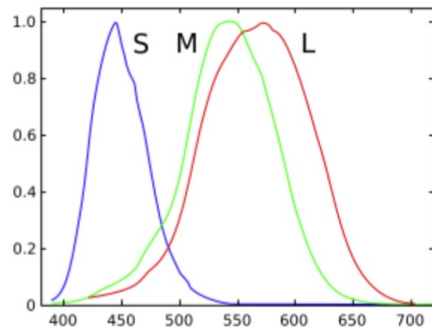
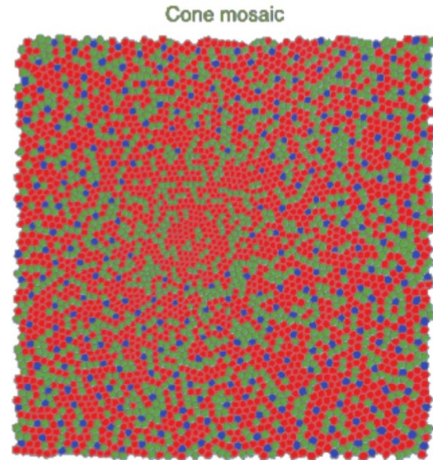


Spectra of some real-world surfaces





Color sensing in human vs. camera





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Quantifying color perception

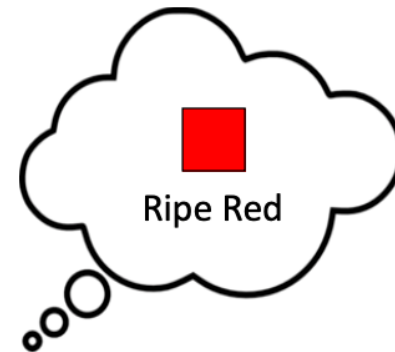


- Spectral distributions go through a “black box” (human visual system) and are perceived as color.
- The only way to quantify the “black box” is to perform a human study.

Spectral distribution



“Black box”



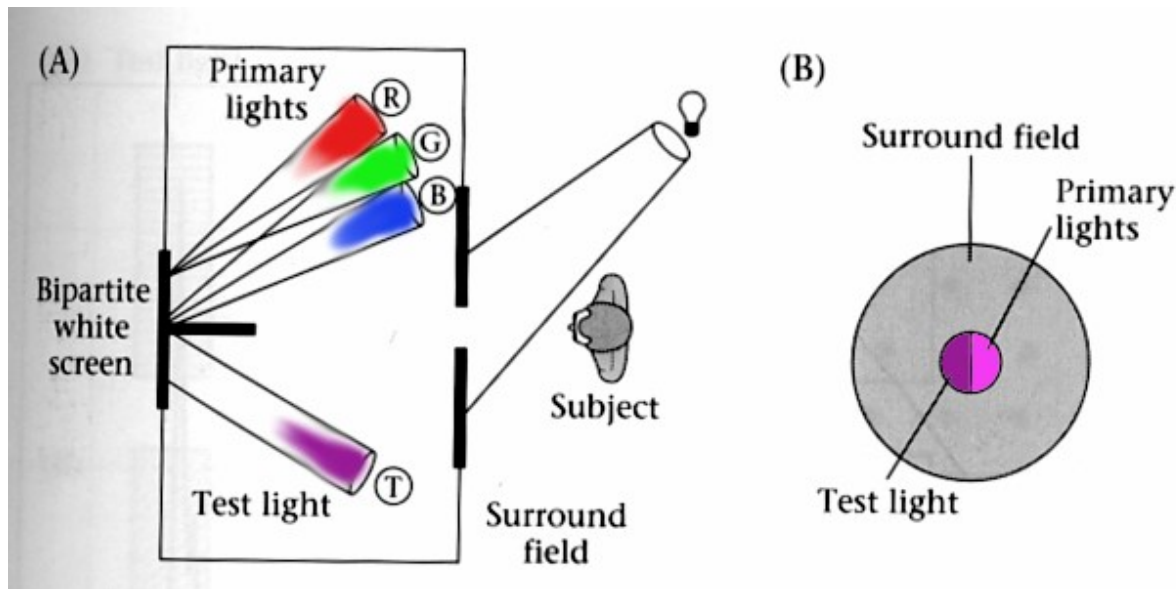
Source: [M. Brown](#)



Color matching experiments



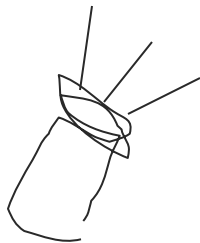
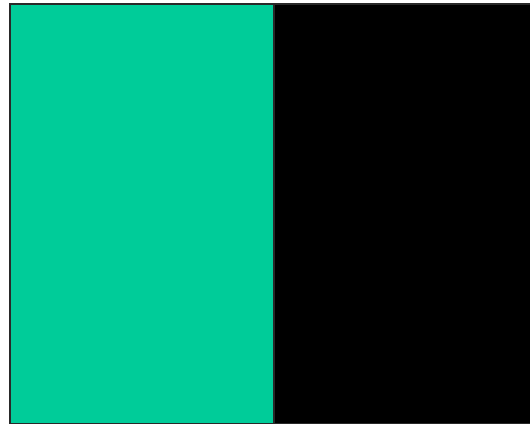
- We would like to understand which spectra produce the same color sensation in people under similar viewing conditions



Wandell, Foundations of Vision, 1995



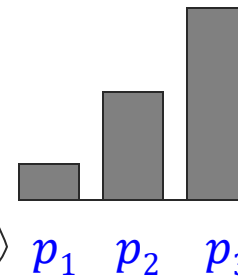
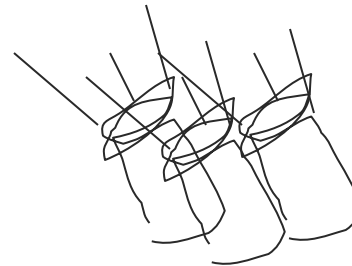
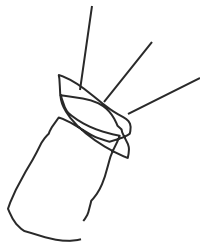
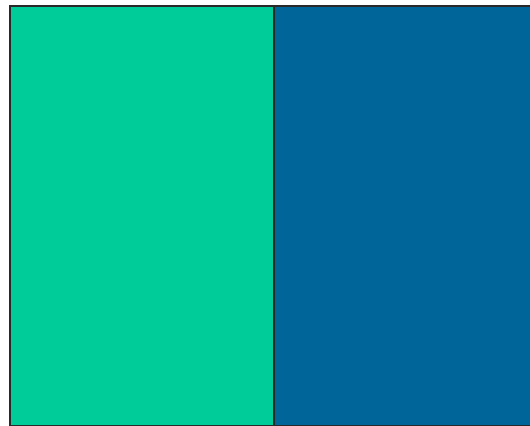
Color matching experiment 1



Source: W. Freeman



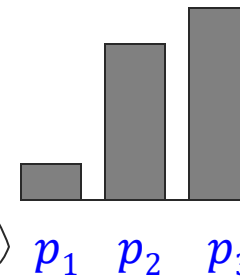
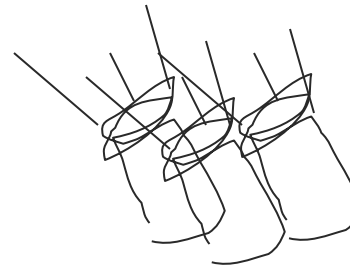
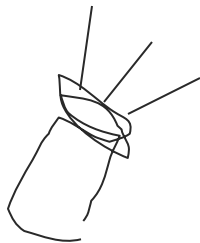
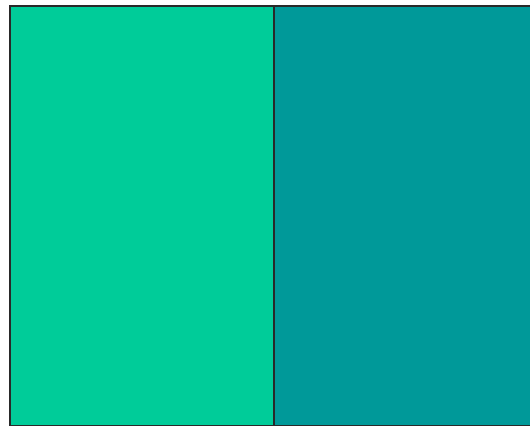
Color matching experiment 1



Source: W. Freeman



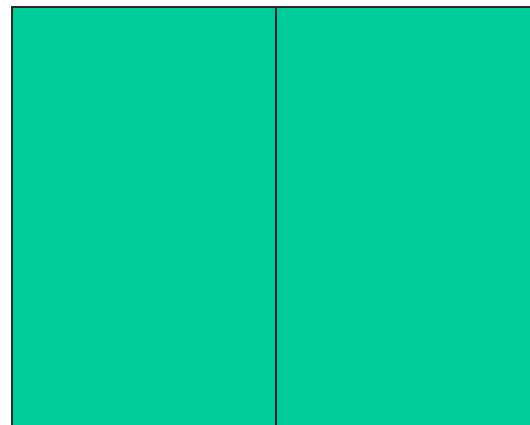
Color matching experiment 1



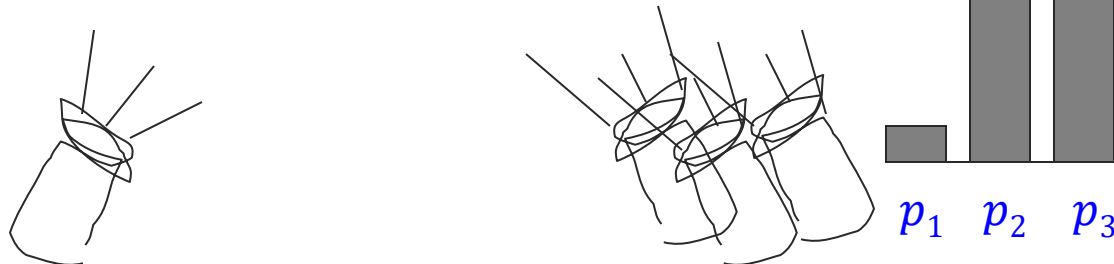
Source: W. Freeman



Color matching experiment 1



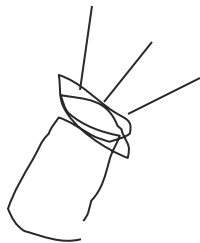
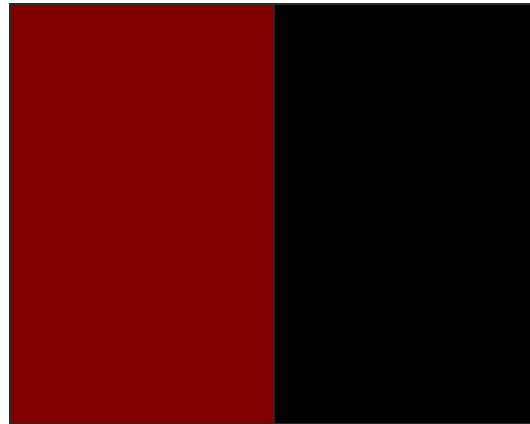
The primary color amounts needed for a match



Source: W. Freeman



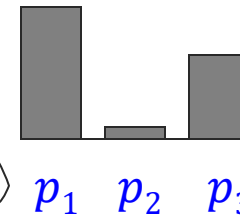
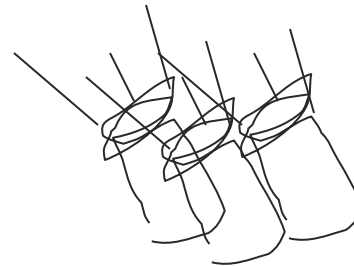
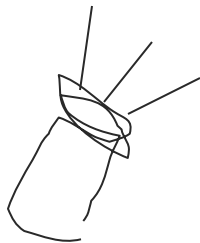
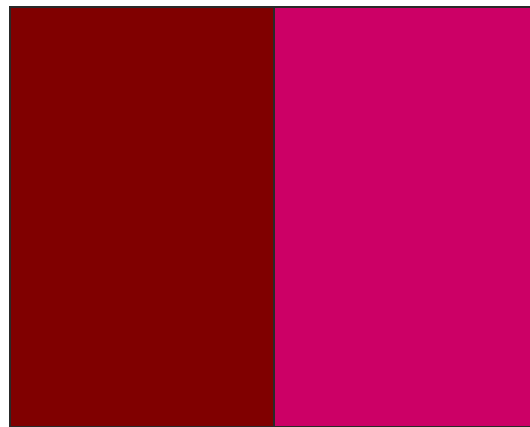
Color matching experiment 2



Source: W. Freeman



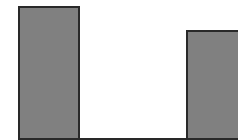
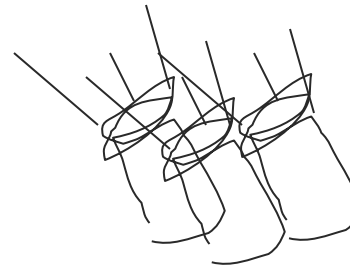
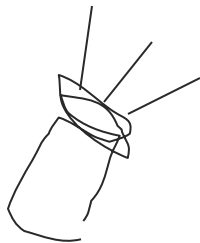
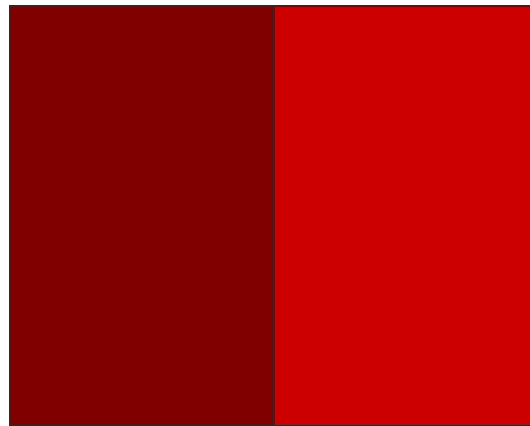
Color matching experiment 2



Source: W. Freeman



Color matching experiment 2



p_1 p_2 p_3

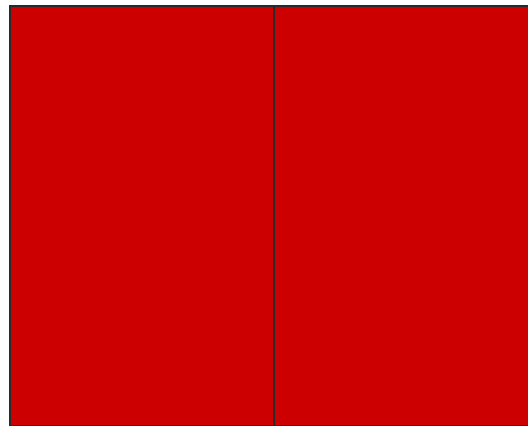
Source: W. Freeman



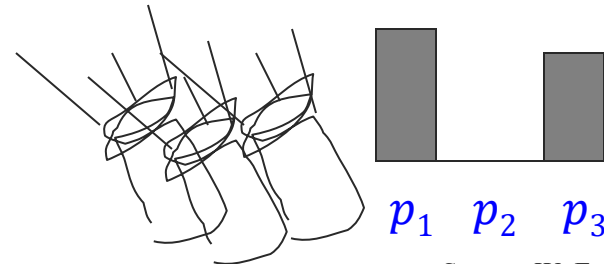
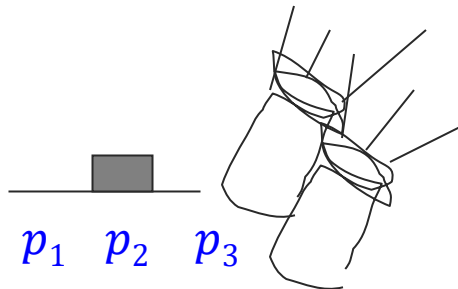
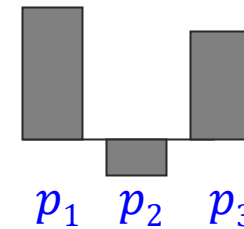
Color matching experiment 2



We say a
“negative”
amount of p_2
was needed to
make the match,
because we
added it to the
test color’s side.



The primary color
amounts needed
for a match:



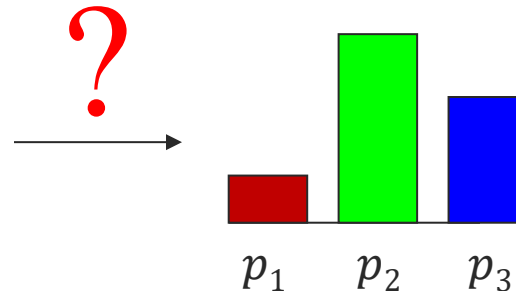
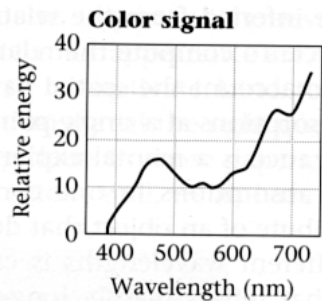
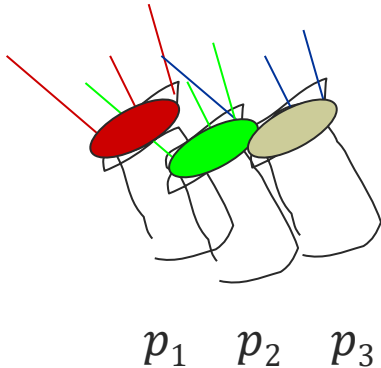
Source: W. Freeman



Linear color spaces



- Fixing three **primaries** defines a linear color space in which the **coordinates** of a color are given by the weights of the primaries used to match it.
- How can we find the coordinates of an arbitrary color signal?
- We need **matching functions**, or amounts of each primary needed to match monochromatic sources at each wavelength

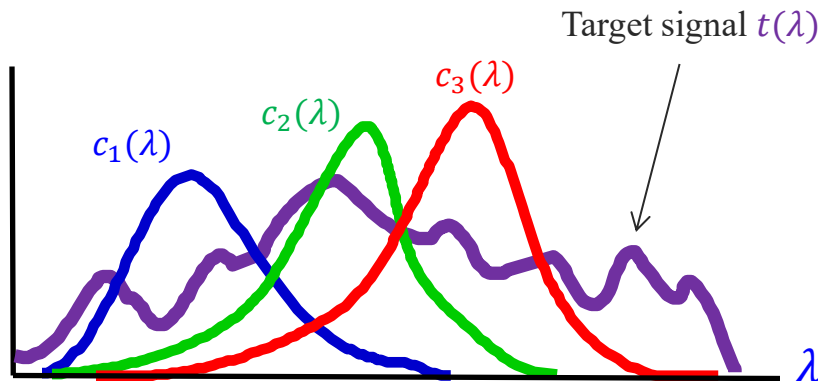




Linear color spaces



- Let $t(\lambda)$ be the spectrum of the target signal and $c_1(\lambda)$, $c_2(\lambda)$, and $c_3(\lambda)$ the matching functions



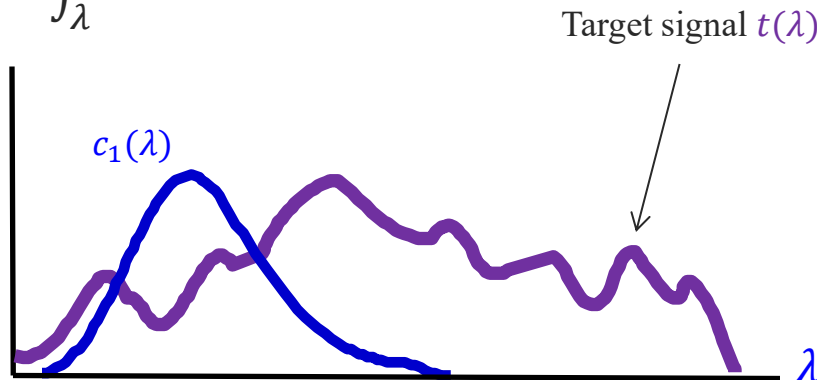


Linear color spaces



- Let $t(\lambda)$ be the spectrum of the target signal and $c_1(\lambda)$, $c_2(\lambda)$, and $c_3(\lambda)$ the matching functions
- Then the weights of the primaries needed to match $t(\lambda)$ are:

$$w_1 = \int_{\lambda} t(\lambda) c_1(\lambda) d\lambda$$





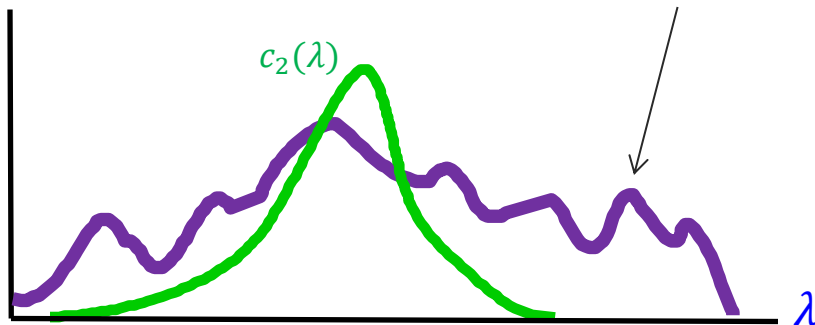
Linear color spaces



- Let $t(\lambda)$ be the spectrum of the target signal and $c_1(\lambda)$, $c_2(\lambda)$, and $c_3(\lambda)$ the matching functions
- Then the weights of the primaries needed to match $t(\lambda)$ are:

$$w_1 = \int_{\lambda} t(\lambda) c_1(\lambda) d\lambda \quad w_2 = \int_{\lambda} t(\lambda) c_2(\lambda) d\lambda$$

Target signal $t(\lambda)$



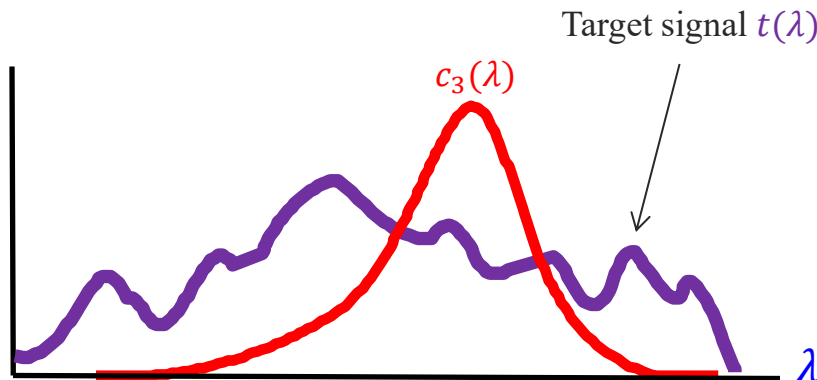


Linear color spaces



- Let $t(\lambda)$ be the spectrum of the target signal and $c_1(\lambda)$, $c_2(\lambda)$, and $c_3(\lambda)$ the matching functions
- Then the weights of the primaries needed to match $t(\lambda)$ are:

$$w_1 = \int_{\lambda} t(\lambda) c_1(\lambda) d\lambda \quad w_2 = \int_{\lambda} t(\lambda) c_2(\lambda) d\lambda \quad w_3 = \int_{\lambda} t(\lambda) c_3(\lambda) d\lambda$$



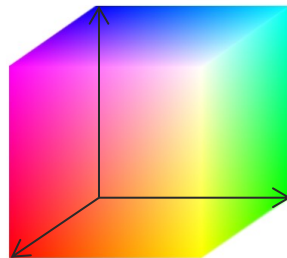
Matching functions act as filters on the target spectrum, like response curves of color receptors!



RGB color space

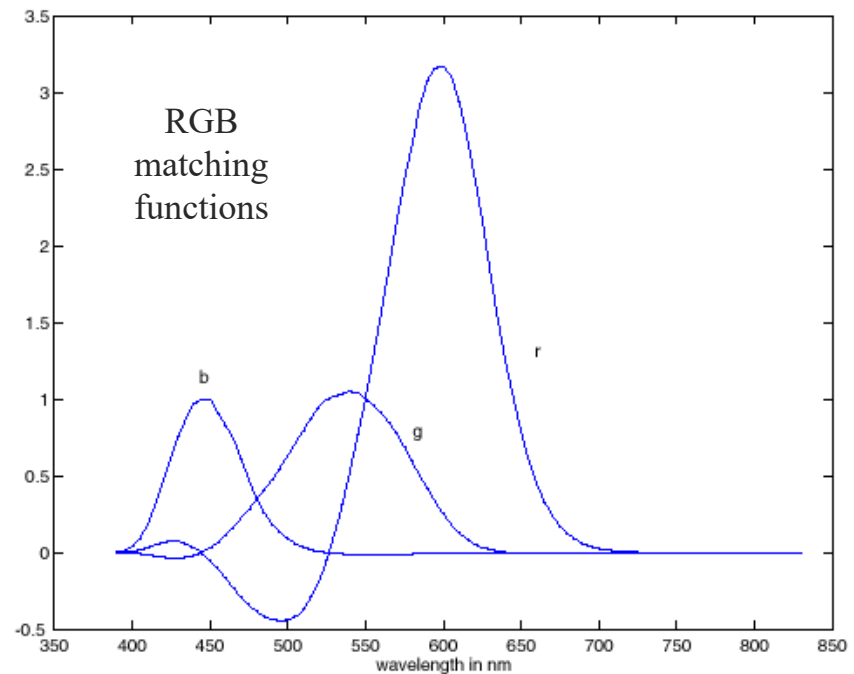


- Primaries are single-wavelength sources, matching functions for R and G have negative values for parts of the spectrum



RGB primaries

- $p_1 = 645.2 \text{ nm}$
- $p_2 = 525.3 \text{ nm}$
- $p_3 = 444.4 \text{ nm}$





Linear color spaces: CIE XYZ



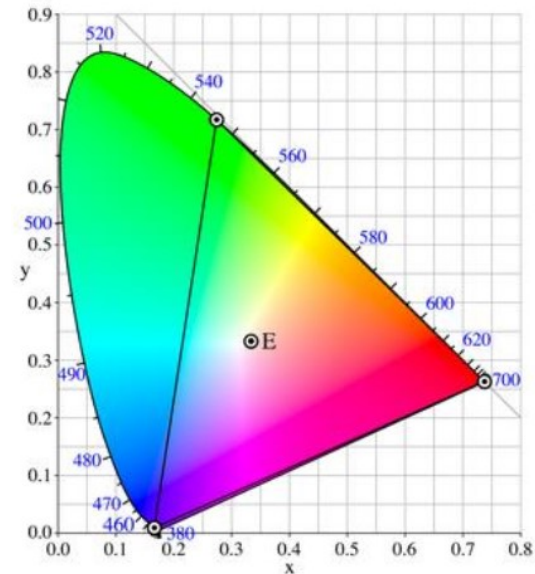
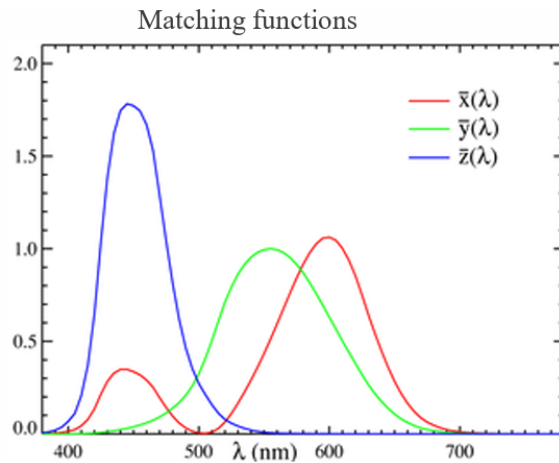
- Primaries are *imaginary*, but matching functions are everywhere positive
- The *Y* parameter corresponds to brightness or *luminance*

Matching functions

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \frac{1}{0.17697} \begin{bmatrix} 0.49 & 0.31 & 0.20 \\ 0.17697 & 0.81240 & 0.01063 \\ 0.00 & 0.01 & 0.99 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

2D visualization:

$$x = X/(X + Y + Z),$$
$$y = Y/(X + Y + Z)$$



http://en.wikipedia.org/wiki/CIE_1931_color_space

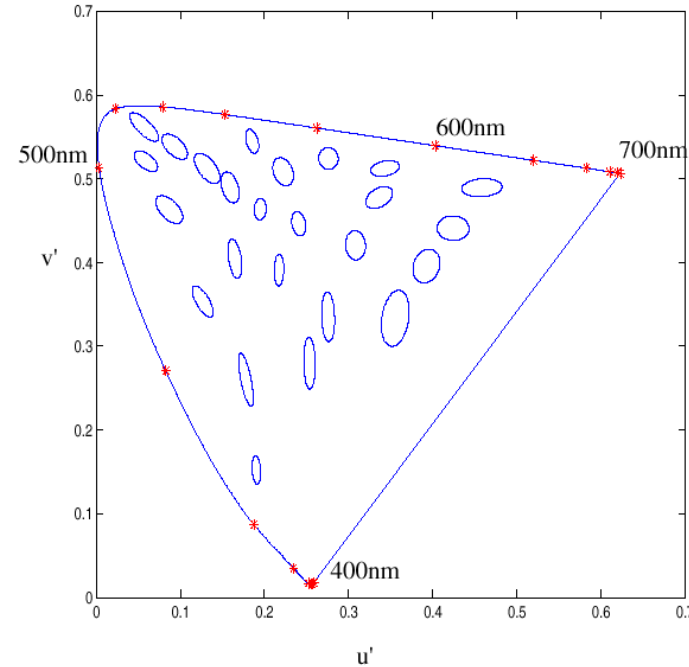
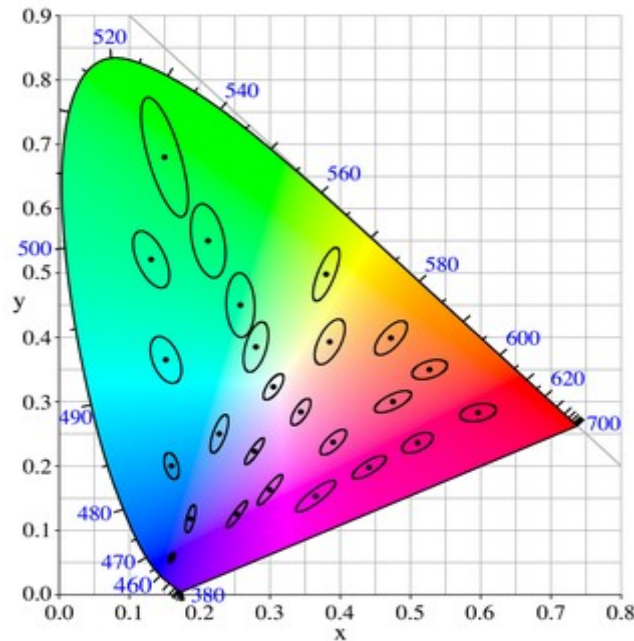
国际照明委员会（International Commission on Illumination, CIE）



Uniform color spaces



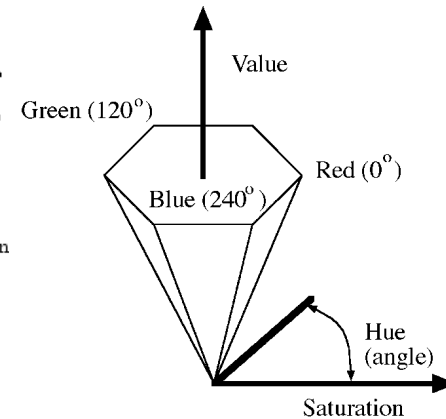
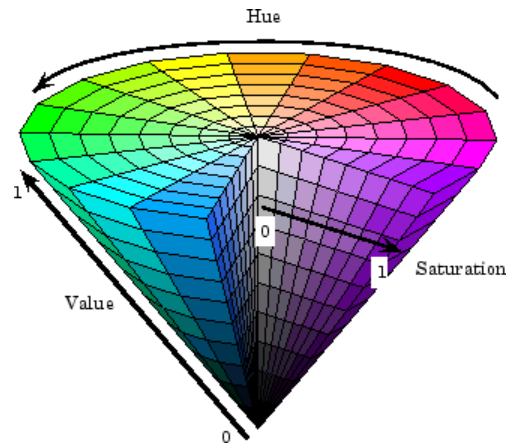
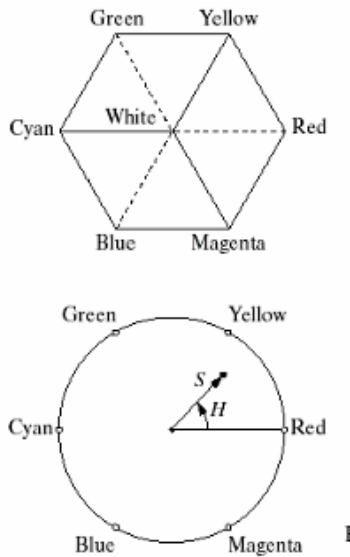
- Unfortunately, differences in x, y coordinates do not reflect perceptual color differences
- CIE $u'v'$ is a projective transform of x, y to make the ellipses more uniform



McAdam ellipses: Just noticeable differences in color



Nonlinear color spaces: HSV



- Perceptually meaningful dimensions: Hue, Saturation, Value (Intensity)
- RGB cube on its vertex



Overview



- What is an image?
- Reflectance properties of surfaces
- What is color: Physical origin of color
- What is color: Physiology of color
- Quantifying color perception and color space
- **Color constancy**



Theorem of Perception



Perceived brightness $\sim$ # photons



Theorem of Perception



~~Perceived brightness $\sim$ # photons~~



Color perception



■ Color/lightness constancy (恒常性)

- The ability of the human visual system to perceive the intrinsic reflectance properties of the surfaces despite changes in illumination conditions



J. S. Sargent, The Daughters of Edward D. Boit, 1882



What color is the dress?



当你看见这张照片时，你的大脑正在根据日光矫正这种色差。所以如果人们认为光源是蓝色从而忽视蓝色的部分，则他们看到白色和金色；而如果他们忽视金色的部分，则他们看到蓝色和黑色



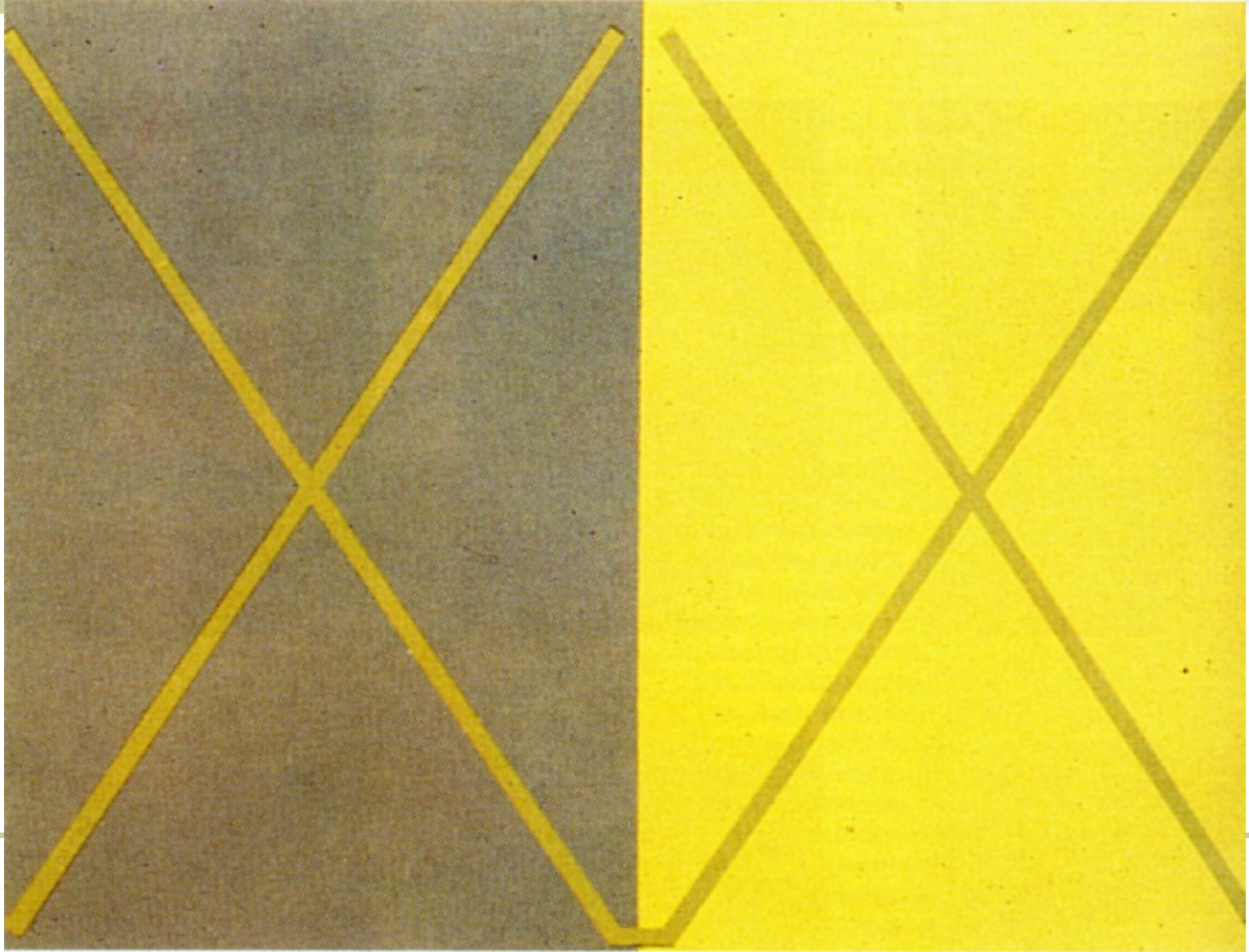
This strawberry cake has no red pixels!

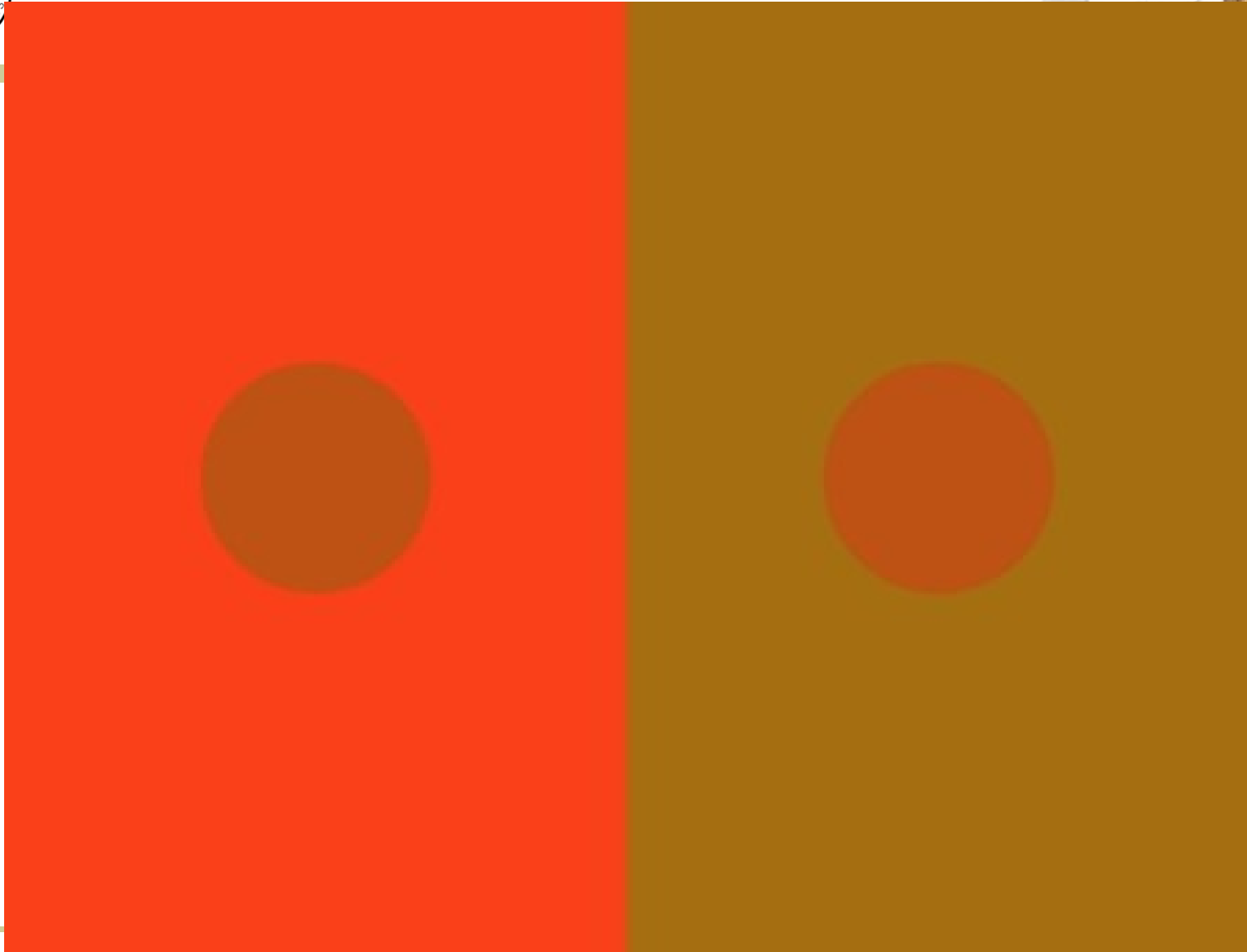


<https://www.digitaltrends.com/photography/non-red-strawberries/>



One source of constancy: local comparisons

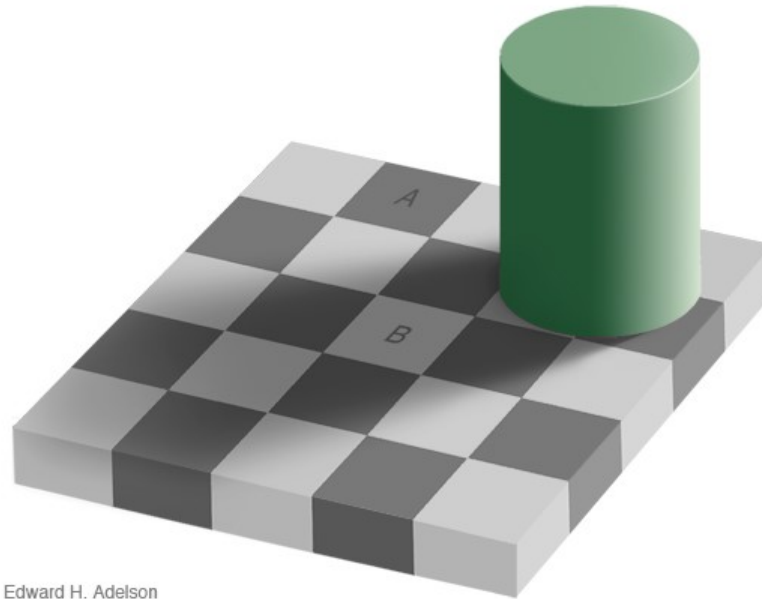






Checker shadow illusion

棋盘阴影错觉



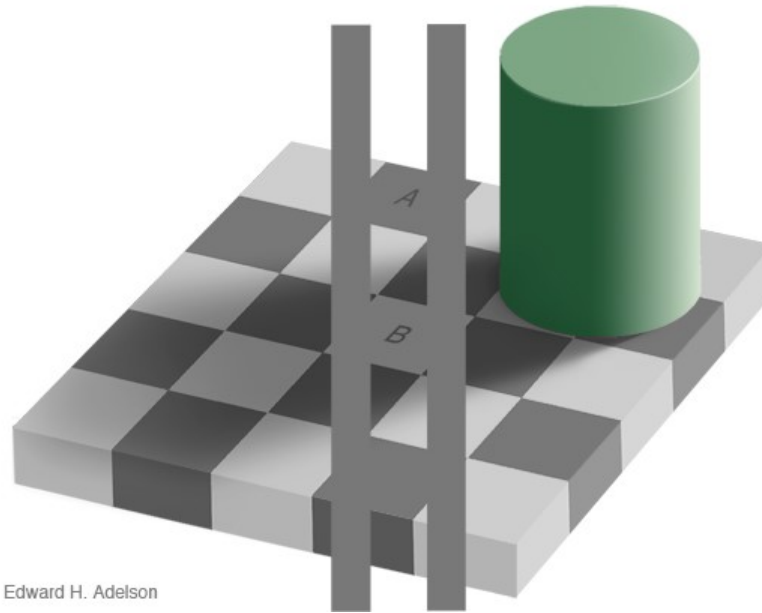
Edward H. Adelson

https://en.wikipedia.org/wiki/Checker_shadow_illusion



Checker shadow illusion

棋盘阴影错觉



Edward H. Adelson

- Possible explanations: simultaneous contrast, reflectance vs. illumination edges

https://en.wikipedia.org/wiki/Checker_shadow_illusion



White balance



- Analogous to color constancy mechanisms in human vision, cameras have mechanisms to adapt to the illumination in the environment **so that neutral (white or gray) objects look neutral.**

Incorrect white balance



Correct white balance



<http://www.cambridgeincolour.com/tutorials/white-balance.htm>



White balance challenges



- When there are several types of illuminants in the scene, different reference points will yield different results



Reference: moon



Reference: stone

<http://www.cambridgeincolour.com/tutorials/white-balance.htm>